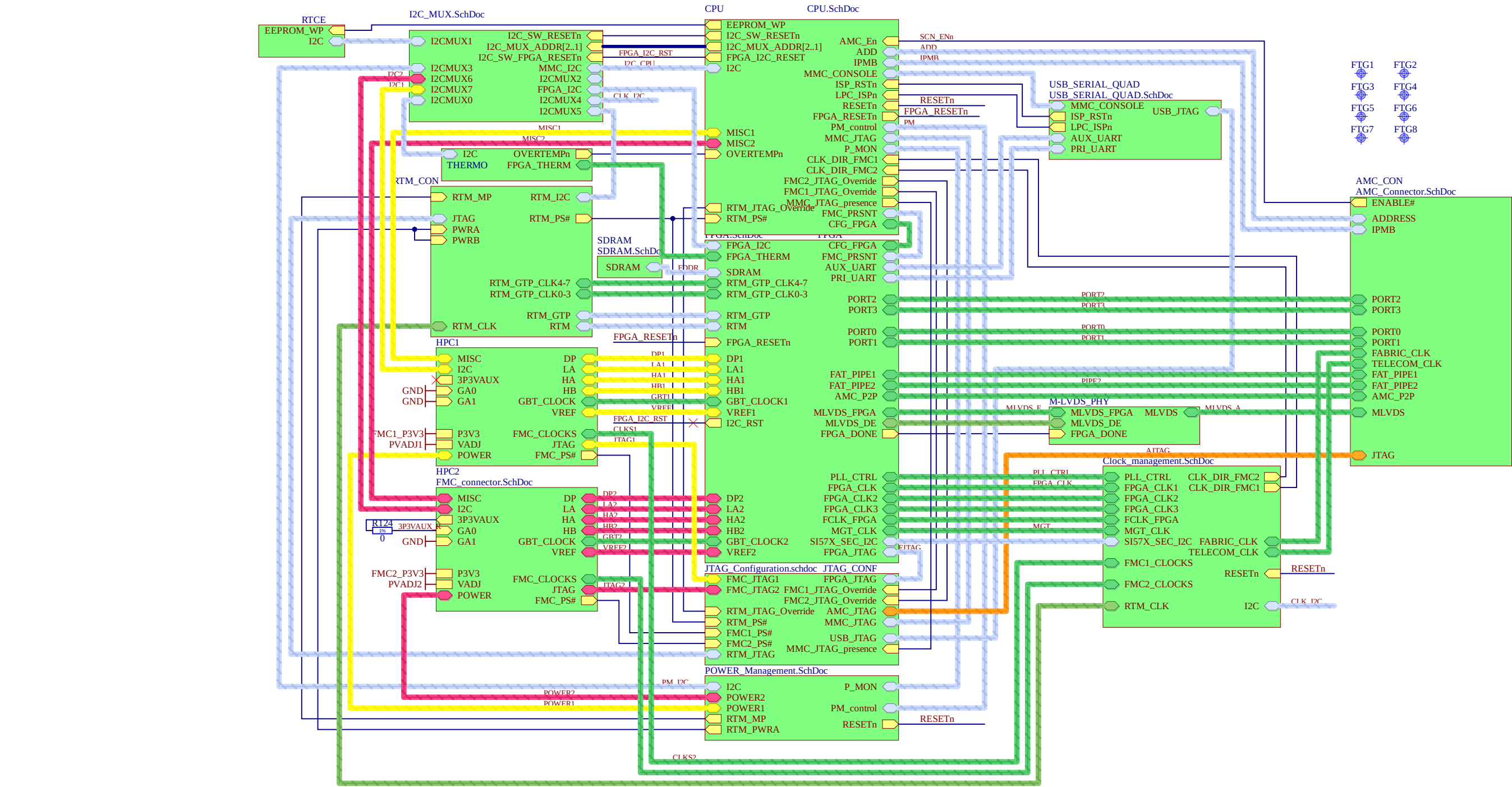




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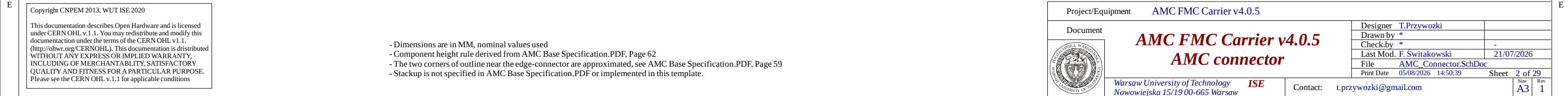


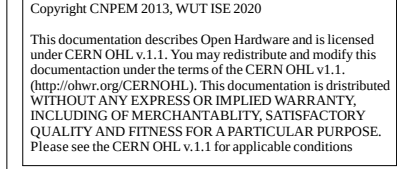
Warsaw University of Technology

Nowowiejska 15/19 00-665 Warsaw

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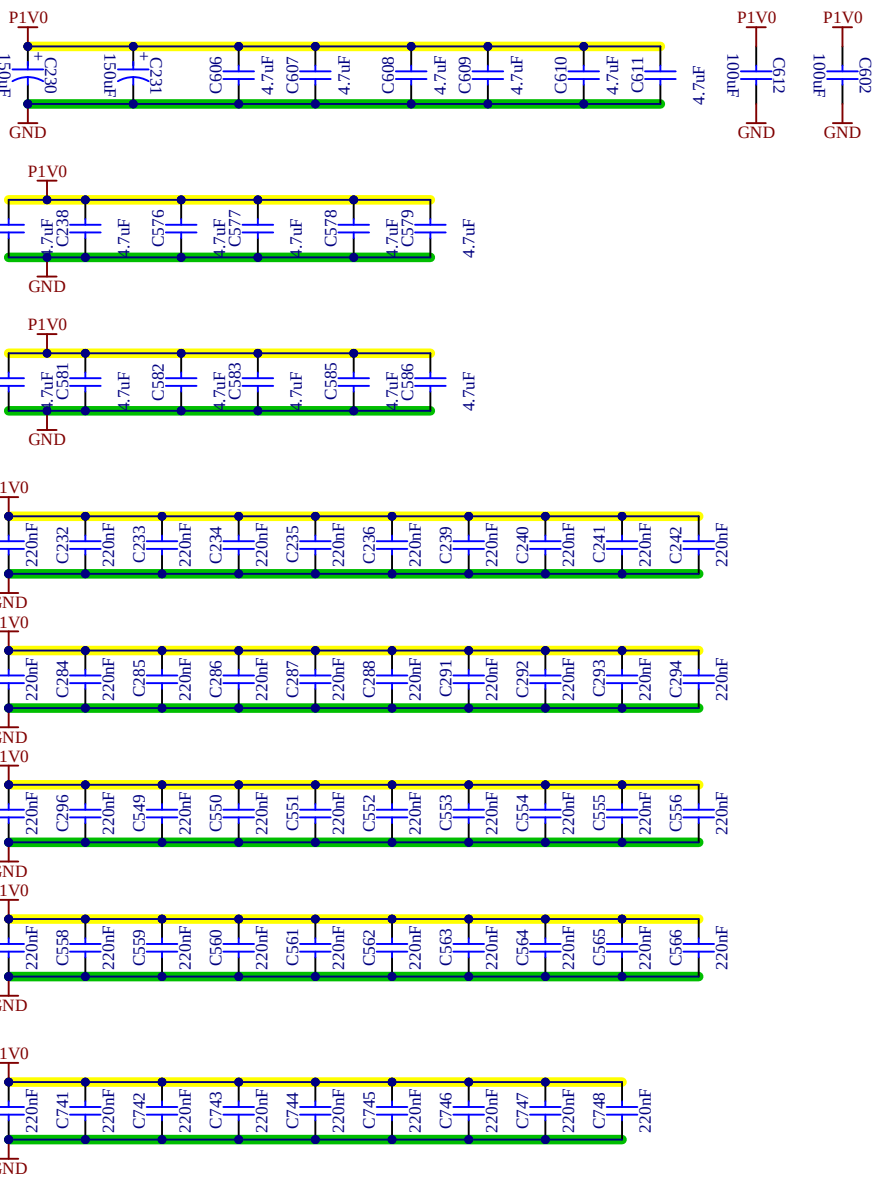
Contact: t.przywozki@gmail.com





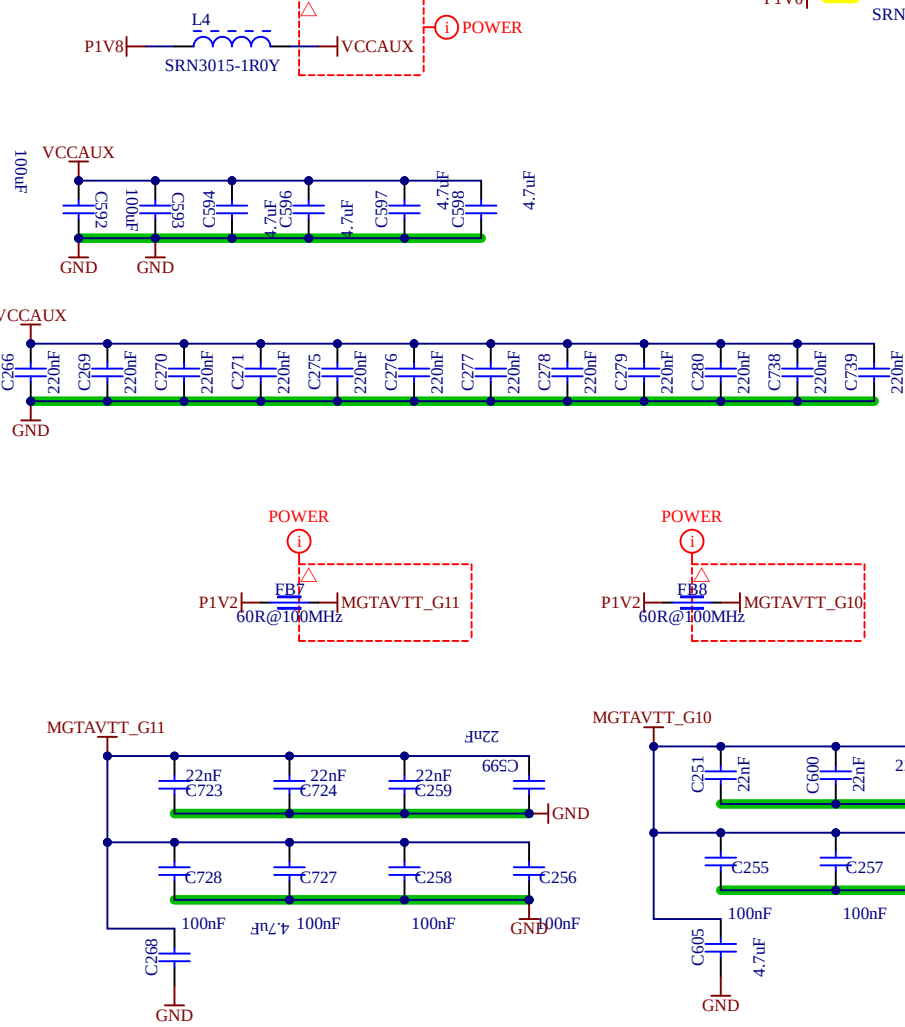
VCCINT

1x680uF, 2x100uF, 28x4.7uF
42x0.47uF on PG.34



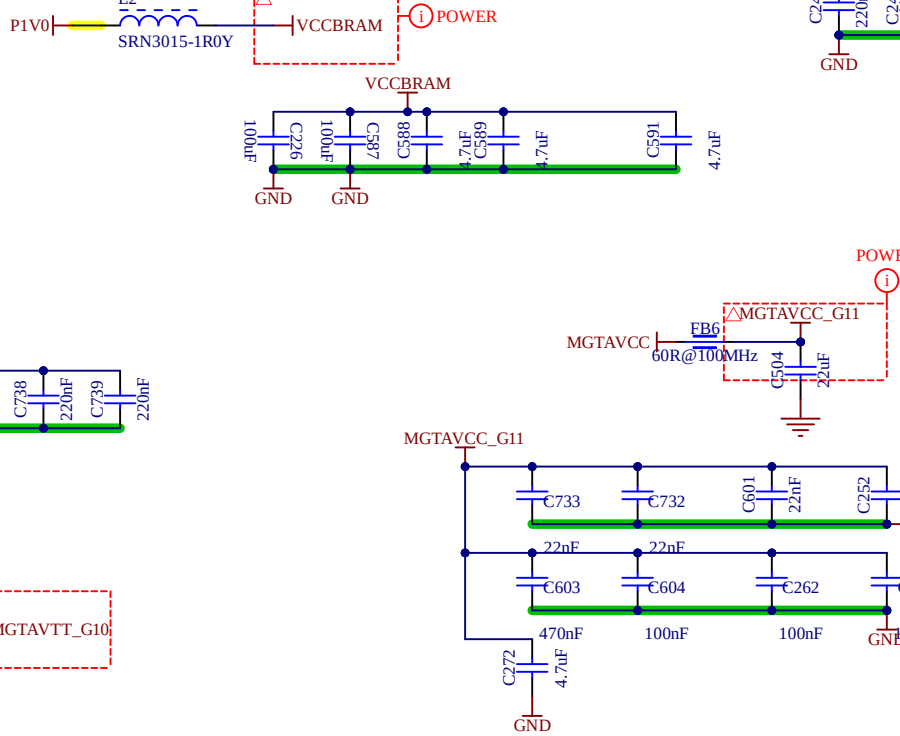
VCCAUX

3x47uF, 4x4.7uF, 7x0.47uF



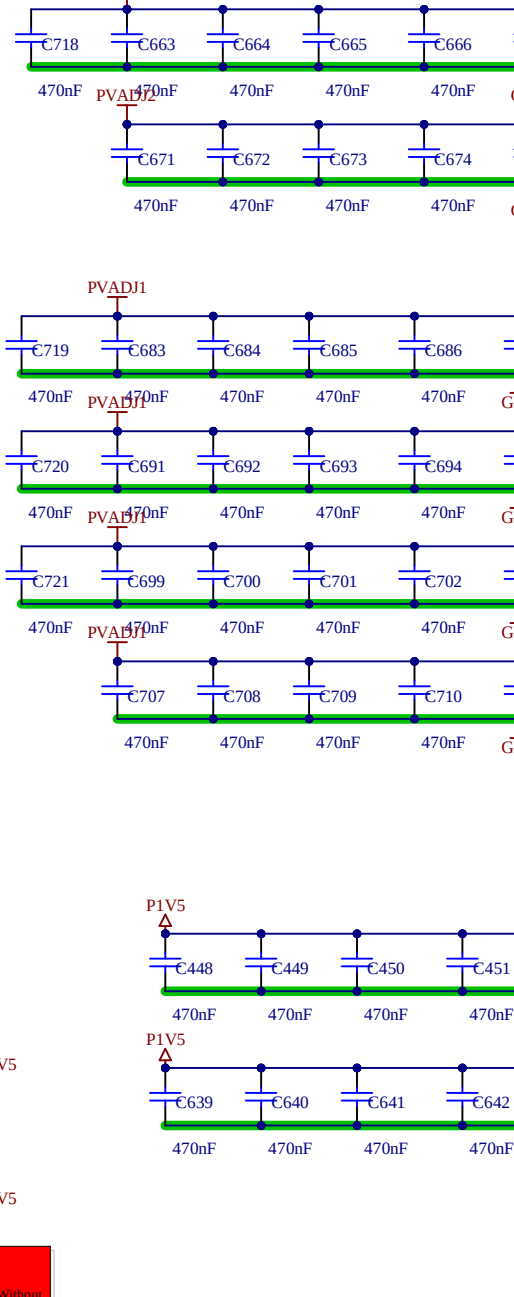
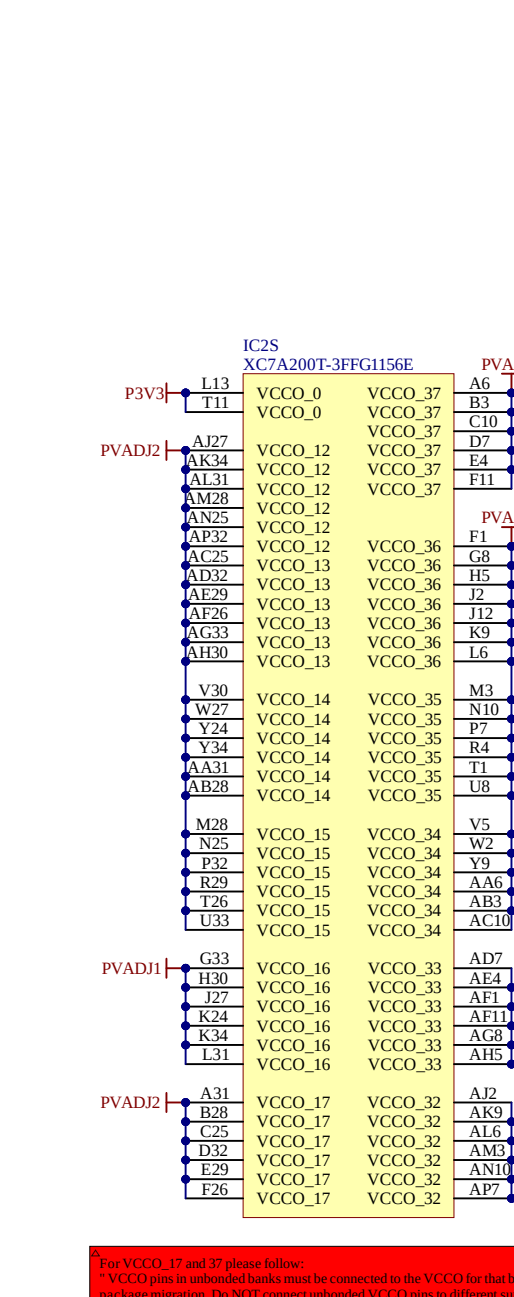
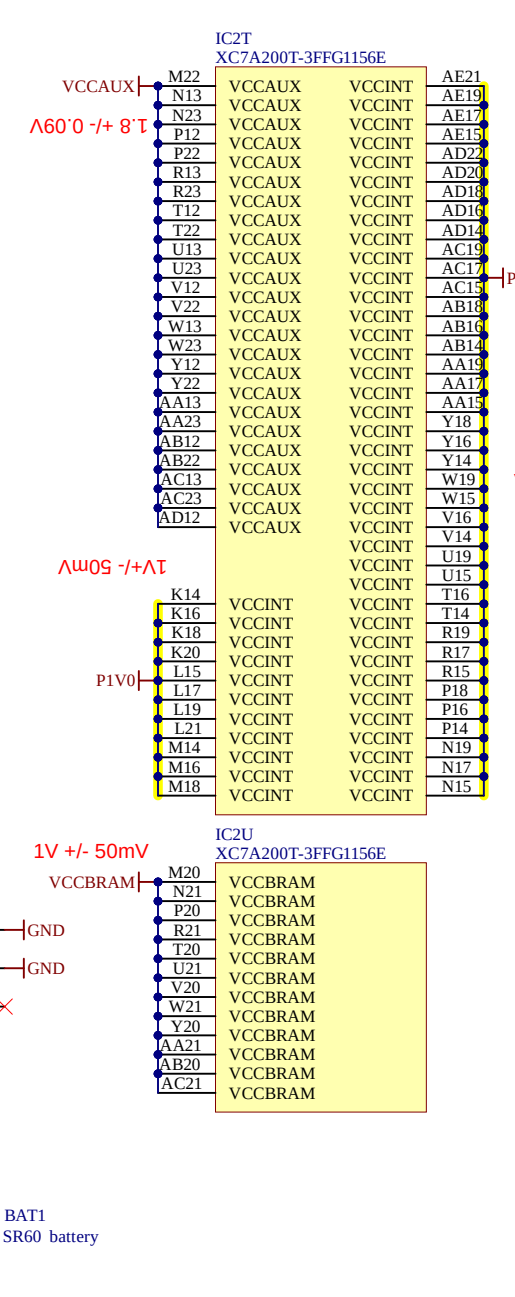
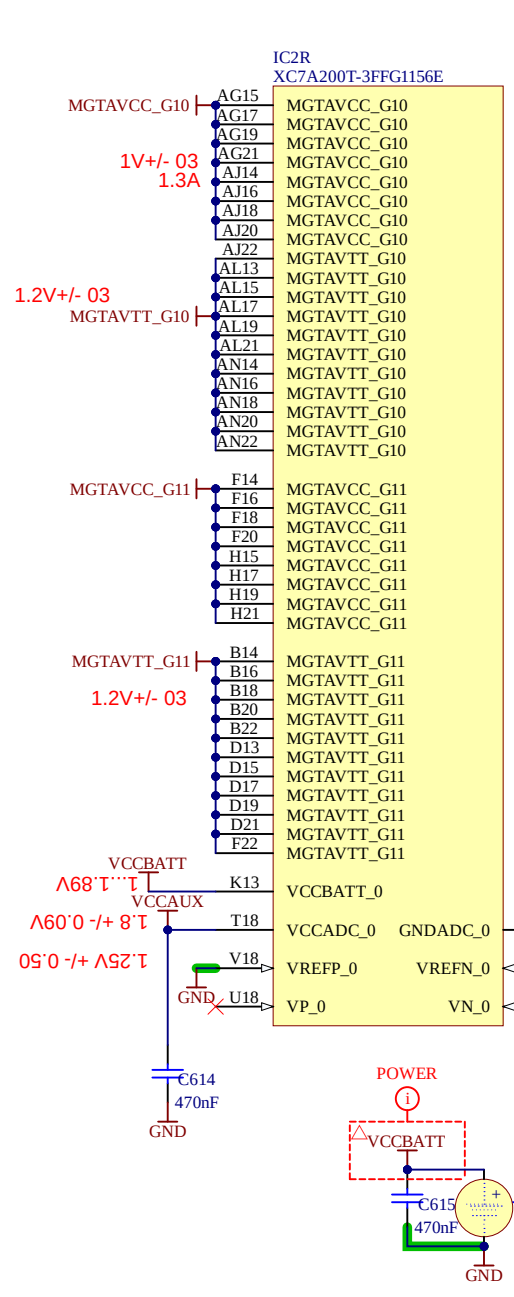
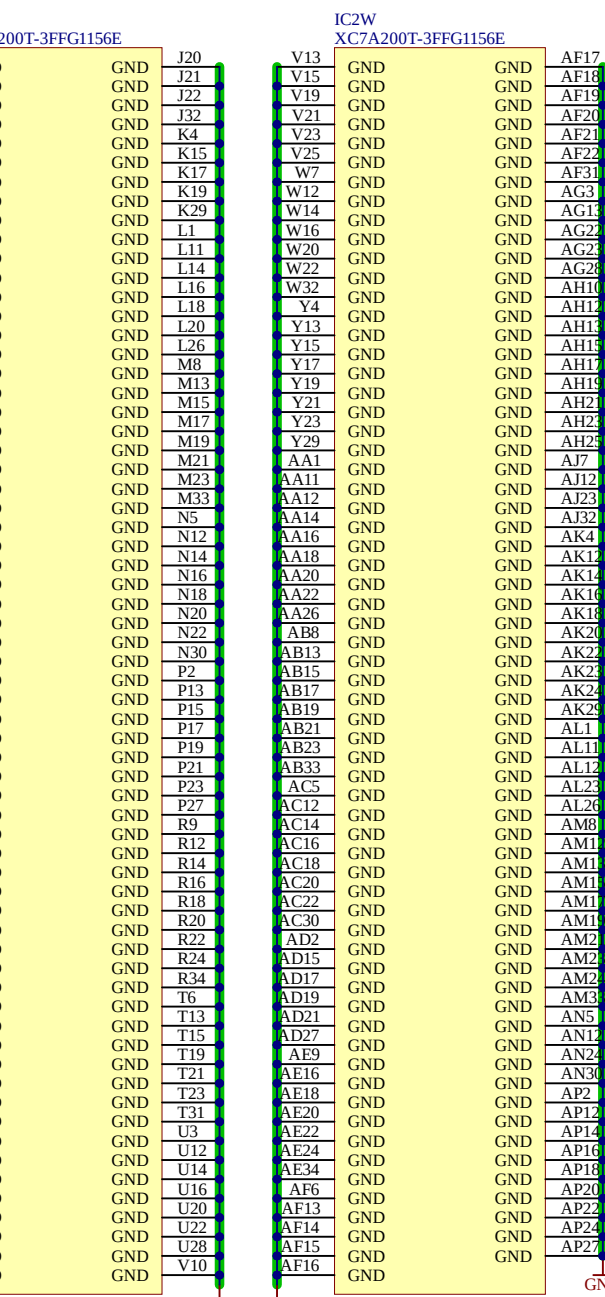
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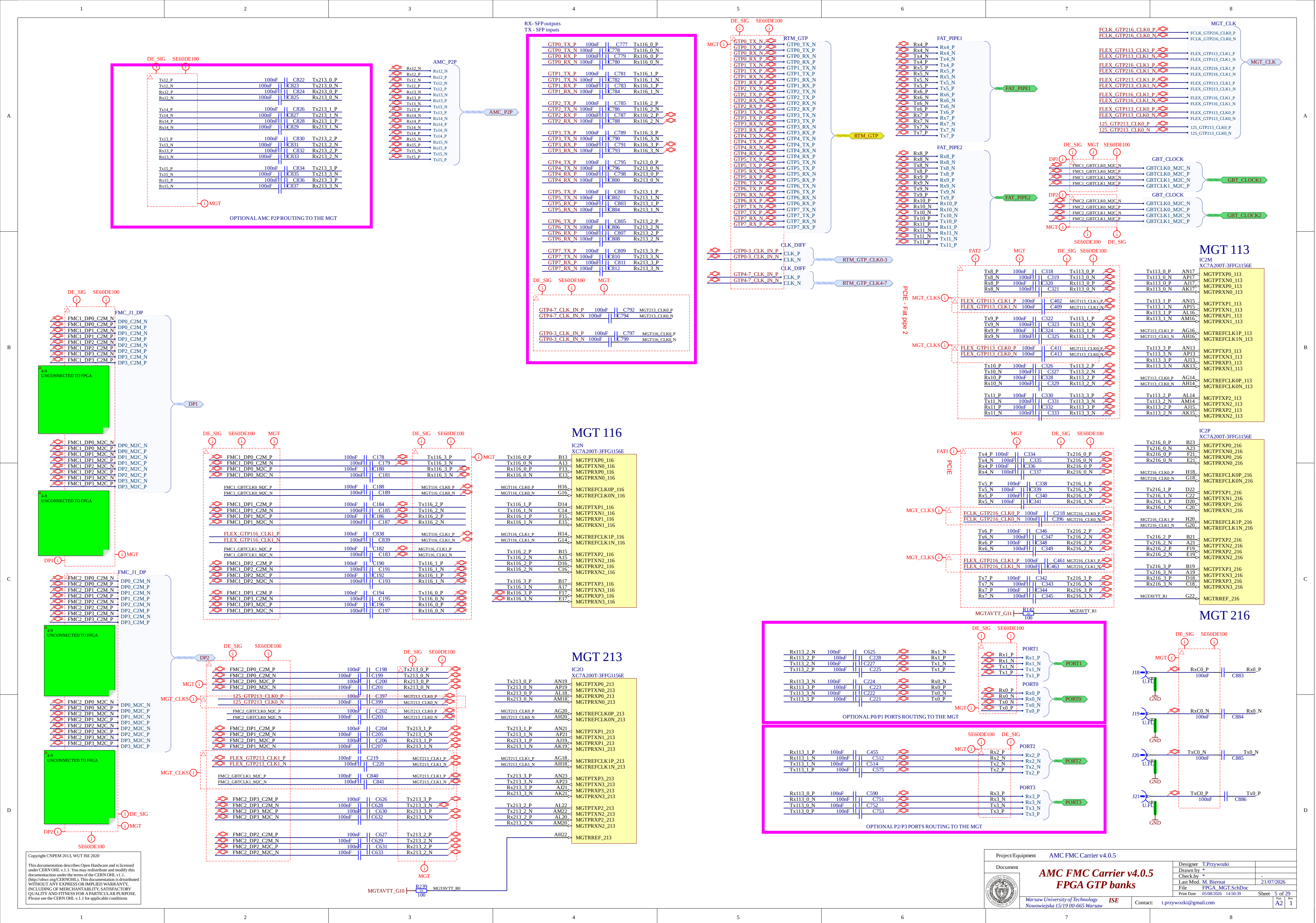
2x100uF, 4x4.7uF, 8x0.47uF



MGTAVCC	1V	2.3A		
MGTAVTT	1.2V	1.5A		
MGTAVCCAUX	1.8V	0.5A		
VCCAUX	1.8V	1A	seq1	
VCCAUX_IO	1.8/2V	0.1A	seq1	
VCCBRAM	1V	1.8A	seq0	
VCCINT	1V	6A	seq0	
VCCO_12	1.2V	3.2A	seq2	
VCCO_135	1.35V	0.9A		seq2
VCCO_15	1.5V	0.9A	seq2	
VCCO_18	1.8V	0.9A	seq2	
VCCO_25	2.5V	0.9A	seq2	
VCCO_33	3.3V	0.9A	seq2	

power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO
voltage difference between VCCO and VCCAUX must not exceed 2.625V for
longer than TVCCO2VCCAUX
power-on sequence to achieve minimum current draw for the GTP transceivers is
VCCINT, VMGTAVCC,
VMGTAVTT or VMGTAVCC, VCCINT, VMGTAVTT. There is no
recommended sequencing for VMGTAVCCAUX. Both VMGTAVCC and
VCCINT can be ramped simultaneously. The recommended power-off sequence is
the reverse of the power-on sequence to
achieve minimum current draw.
When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT -
VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the
VMGTAVTT current draw can increase by 460 mA per transceiver during
VMGTAVCC ramp up. The duration of the current
draw can be up to 0.3 x TMGTAVCC (ramp time from GND to 90% of
VMGTAVCC). The reverse is true for power-down.
• When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT
> 150 mV and VCCINT < 0.7V, the VMGTAVTT current
draw can increase by 50 mA per transceiver during VCCINT ramp up. The
duration of the current draw can be up to
0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true
for power-down.





A

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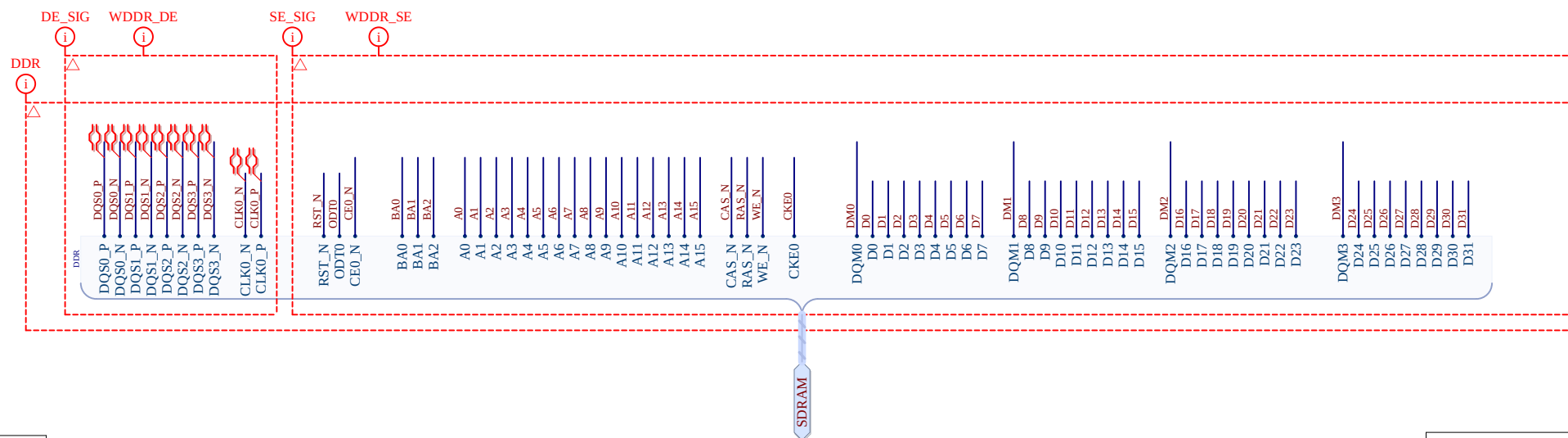
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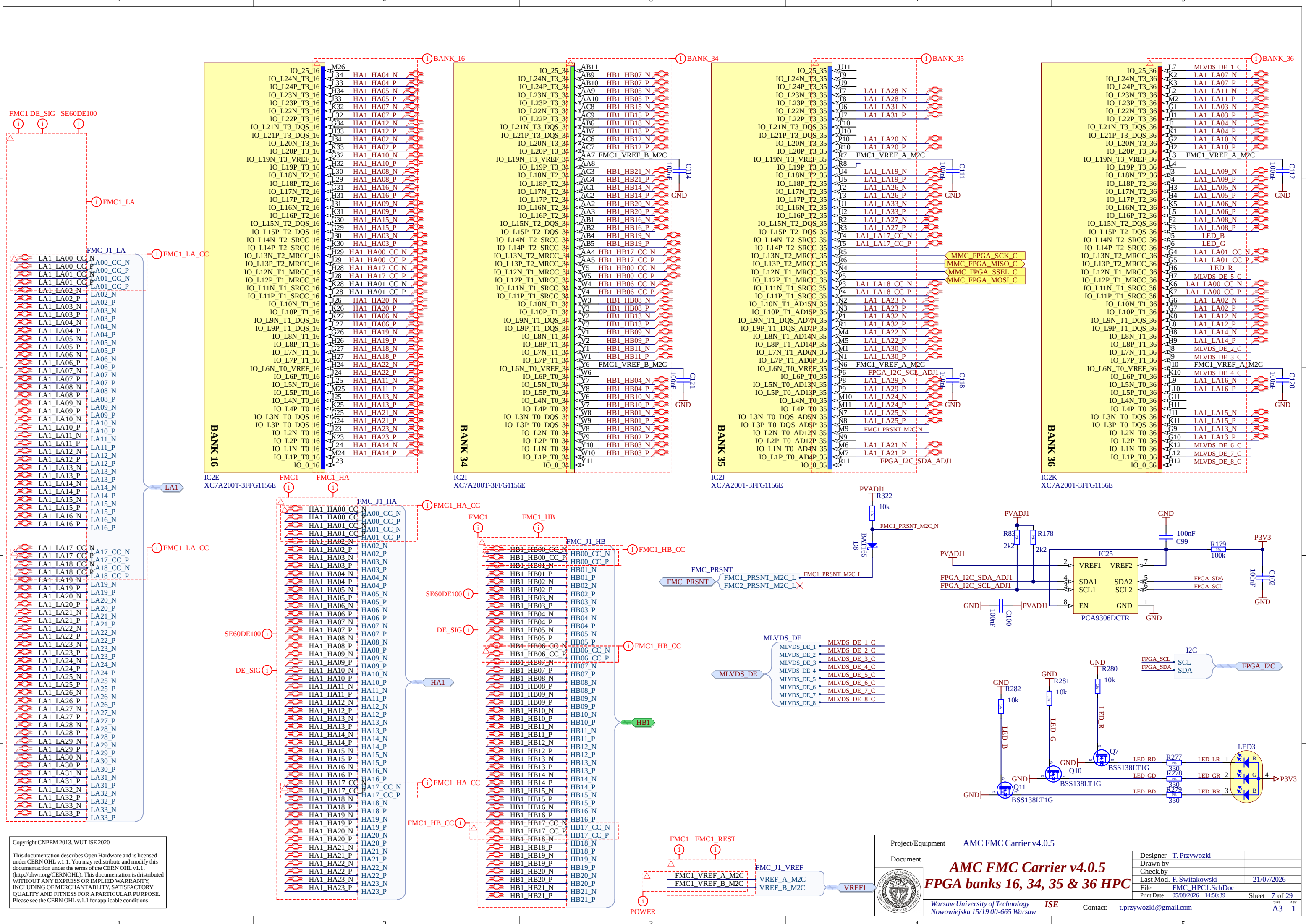
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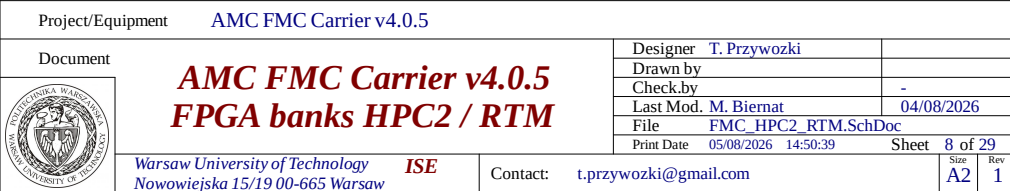
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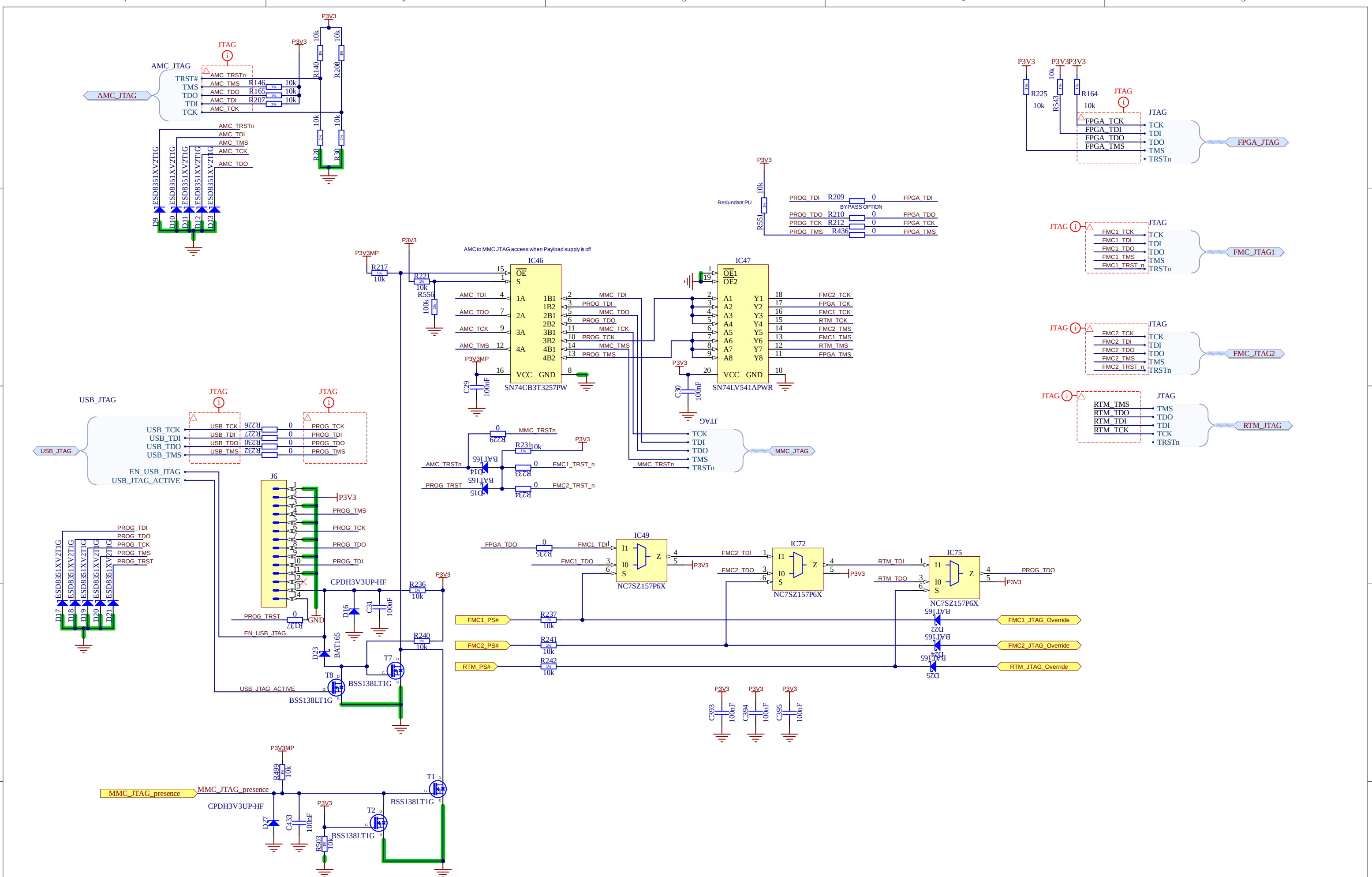
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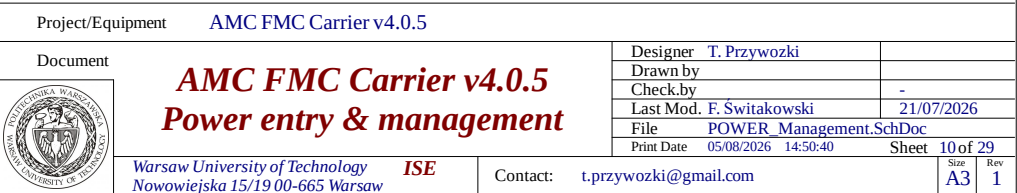
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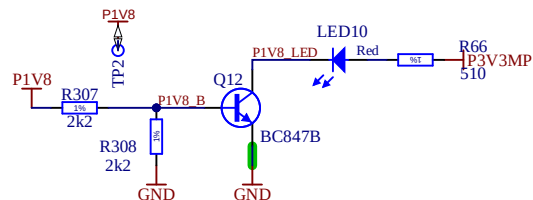
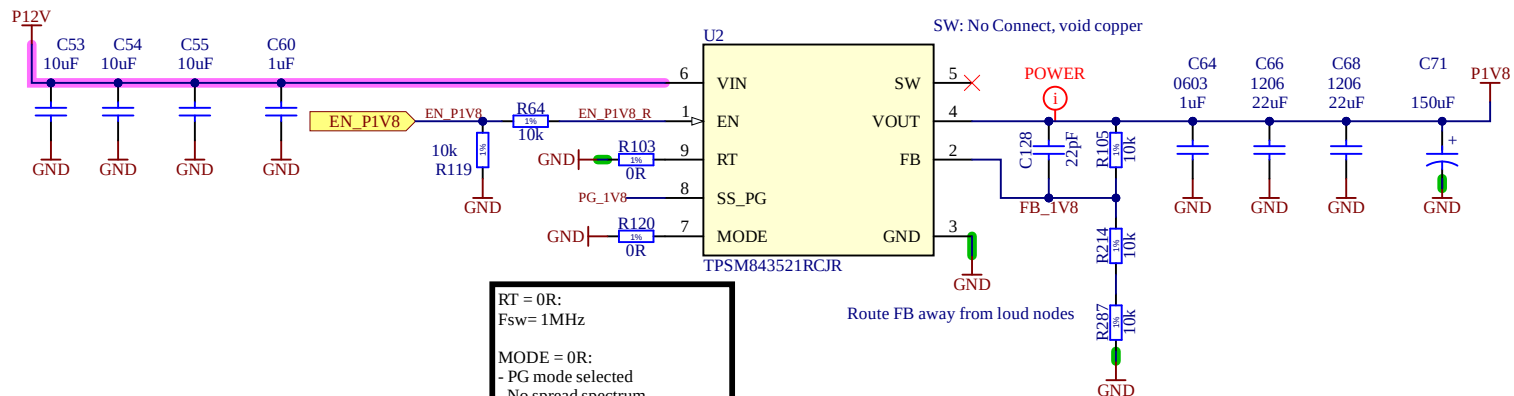
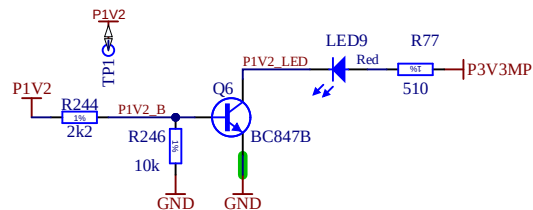
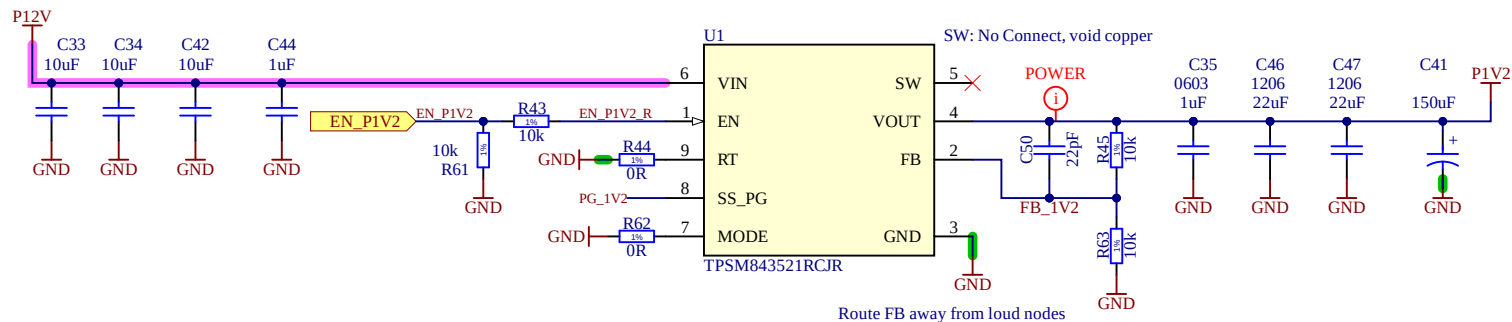








Compatible parts:
TPSM843521 - 5A
TPSM843321 - 3A
TPSM84538 - 5A, 28V in
TPSM84338 - 3A, 28V in

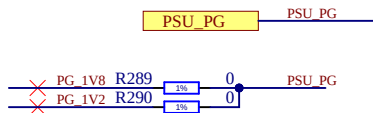


RT = 0R:
Fsw = 1MHz

MODE = 0R:
- PG mode selected
- No spread spectrum
- FCCS mode

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Project/Equipment AMC FMC Carrier v4.0.5

Document



AMC FMC Carrier v4.0.5 Power P1V2 & P1V8

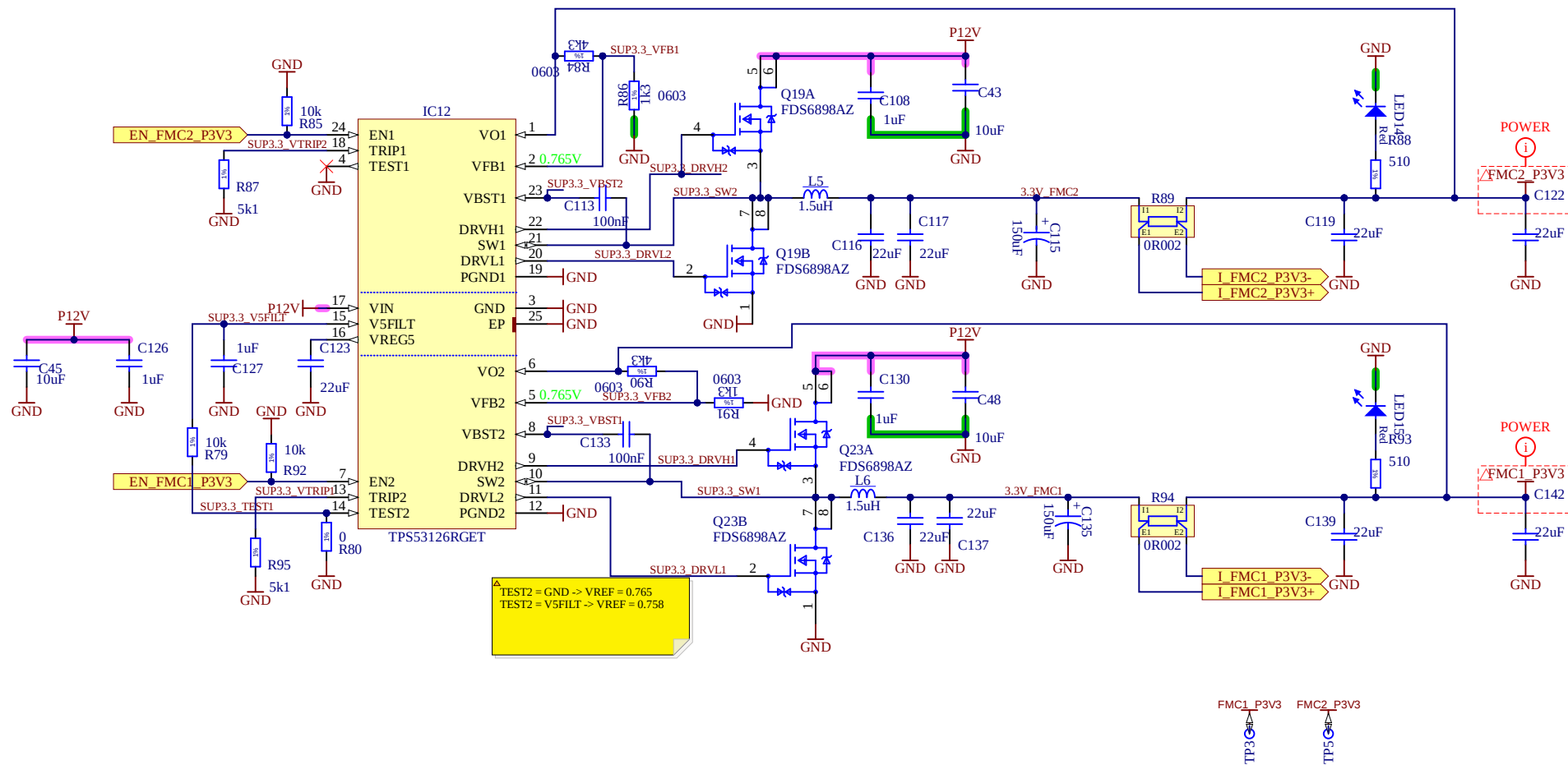
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Contact: t.przywozki@gmail.com

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Last Mod.	F. Świątkowski	21/07/2026
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AMC FMC Carrier v4.0.5 Power FMCs P3V3

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Rev 1

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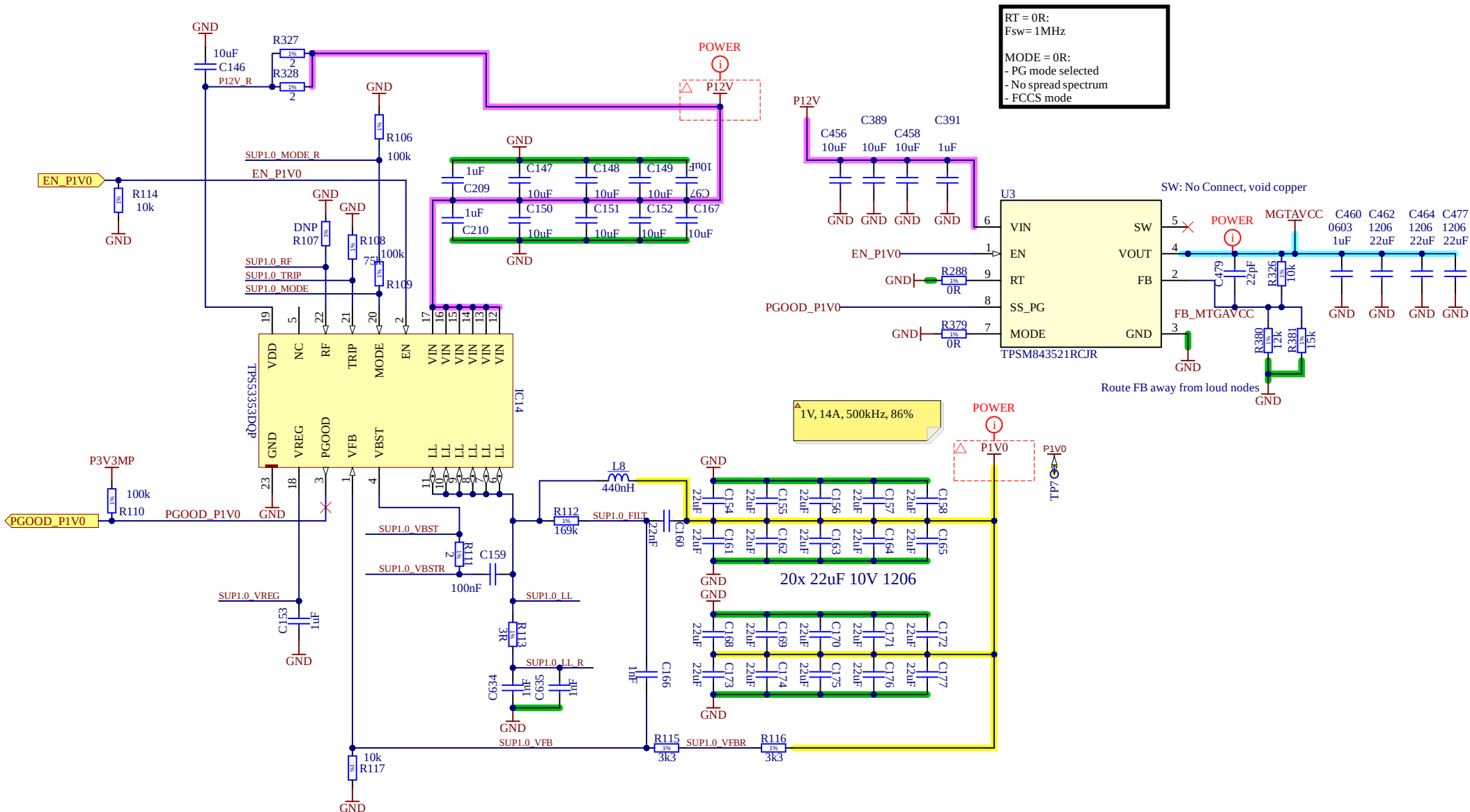
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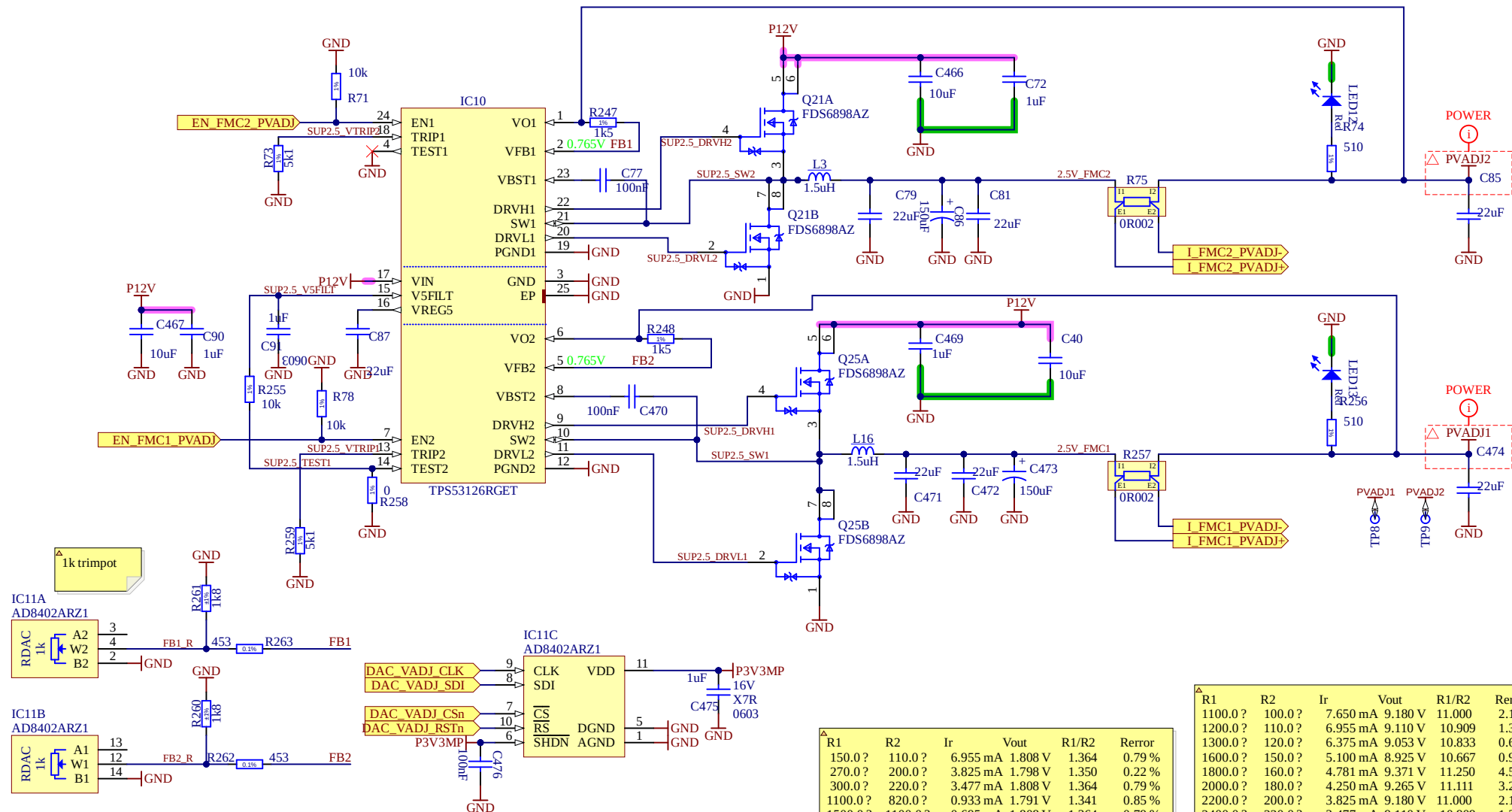
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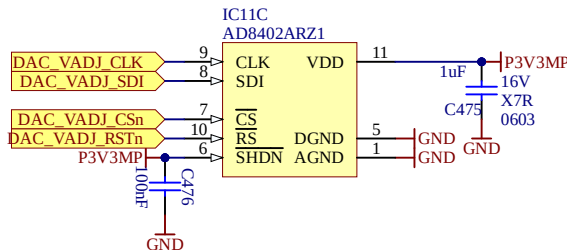
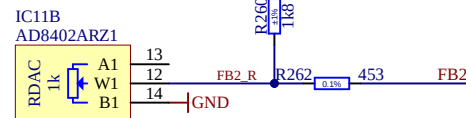
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1k trimpot



R1	R2	I _r	V _{out}	R1/R2	Error
150.0?	110.0?	6.955 mA	1.808 V	1.364	0.79 %
270.0?	200.0?	3.825 mA	1.798 V	1.350	0.22 %
300.0?	220.0?	3.477 mA	1.808 V	1.364	0.79 %
1100.0?	820.0?	0.933 mA	1.791 V	1.341	0.85 %
1500.0?	1100.0?	0.695 mA	1.808 V	1.364	0.79 %
2700.0?	2000.0?	0.383 mA	1.798 V	1.350	0.22 %
3000.0?	2200.0?	0.348 mA	1.808 V	1.364	0.79 %

R1	R2	I _r	V _{out}	R1/R2	Error
1100.0?	100.0?	7.650 mA	9.180 V	11.000	2.19 %
1200.0?	110.0?	6.955 mA	9.110 V	10.909	1.34 %
1300.0?	120.0?	6.375 mA	9.053 V	10.833	0.64 %
1600.0?	150.0?	5.100 mA	8.925 V	10.667	0.91 %
1800.0?	160.0?	4.781 mA	9.371 V	11.250	4.51 %
2000.0?	180.0?	4.250 mA	9.265 V	11.111	3.22 %
2200.0?	200.0?	3.825 mA	9.180 V	11.000	2.19 %
2400.0?	220.0?	3.477 mA	9.110 V	10.909	1.34 %
2700.0?	240.0?	3.188 mA	9.371 V	11.250	4.51 %
3000.0?	270.0?	2.833 mA	9.265 V	11.111	3.22 %
3300.0?	300.0?	2.550 mA	9.180 V	11.000	2.19 %

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Project/Equipment AMC FMC Carrier v4.0.5

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AMC FMC Carrier v4.0.5
Power FMCs PVADJ



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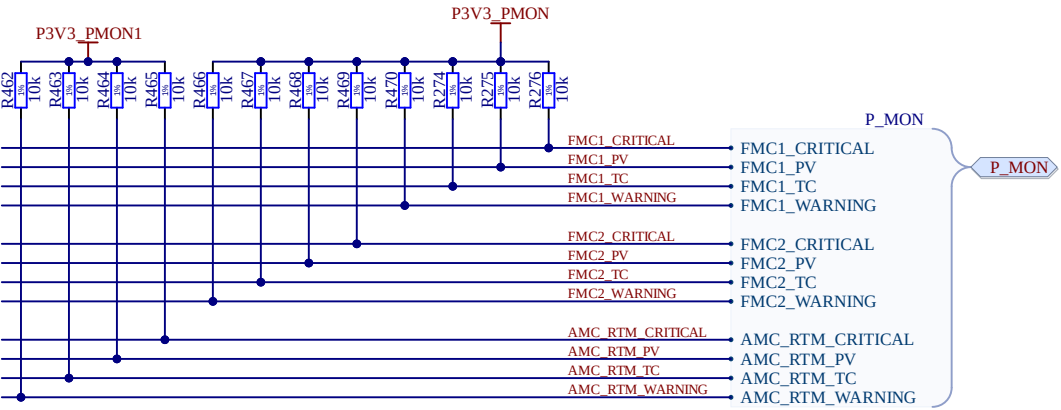
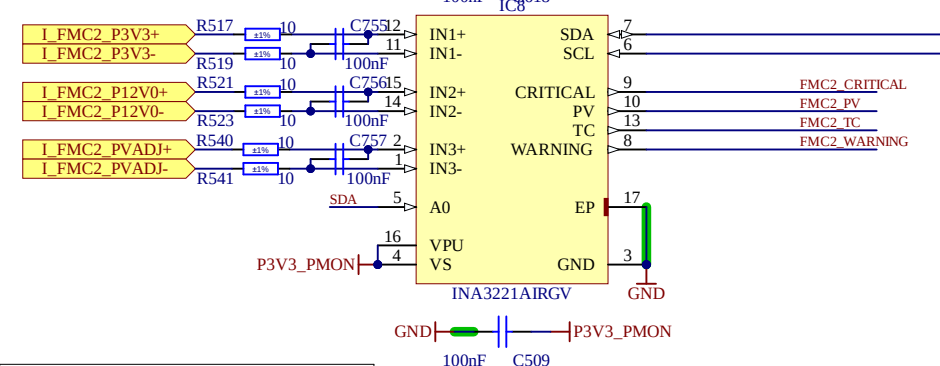
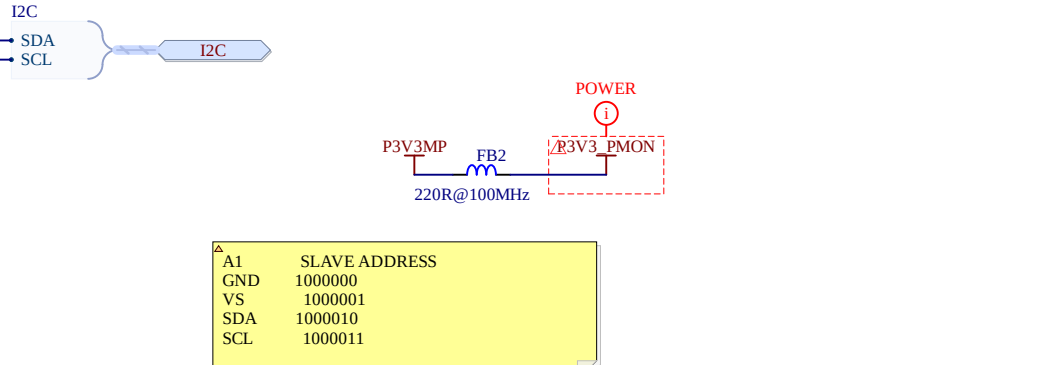
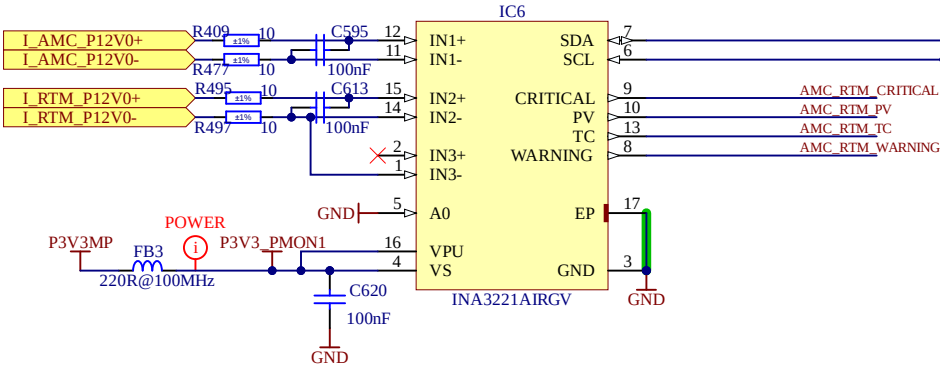
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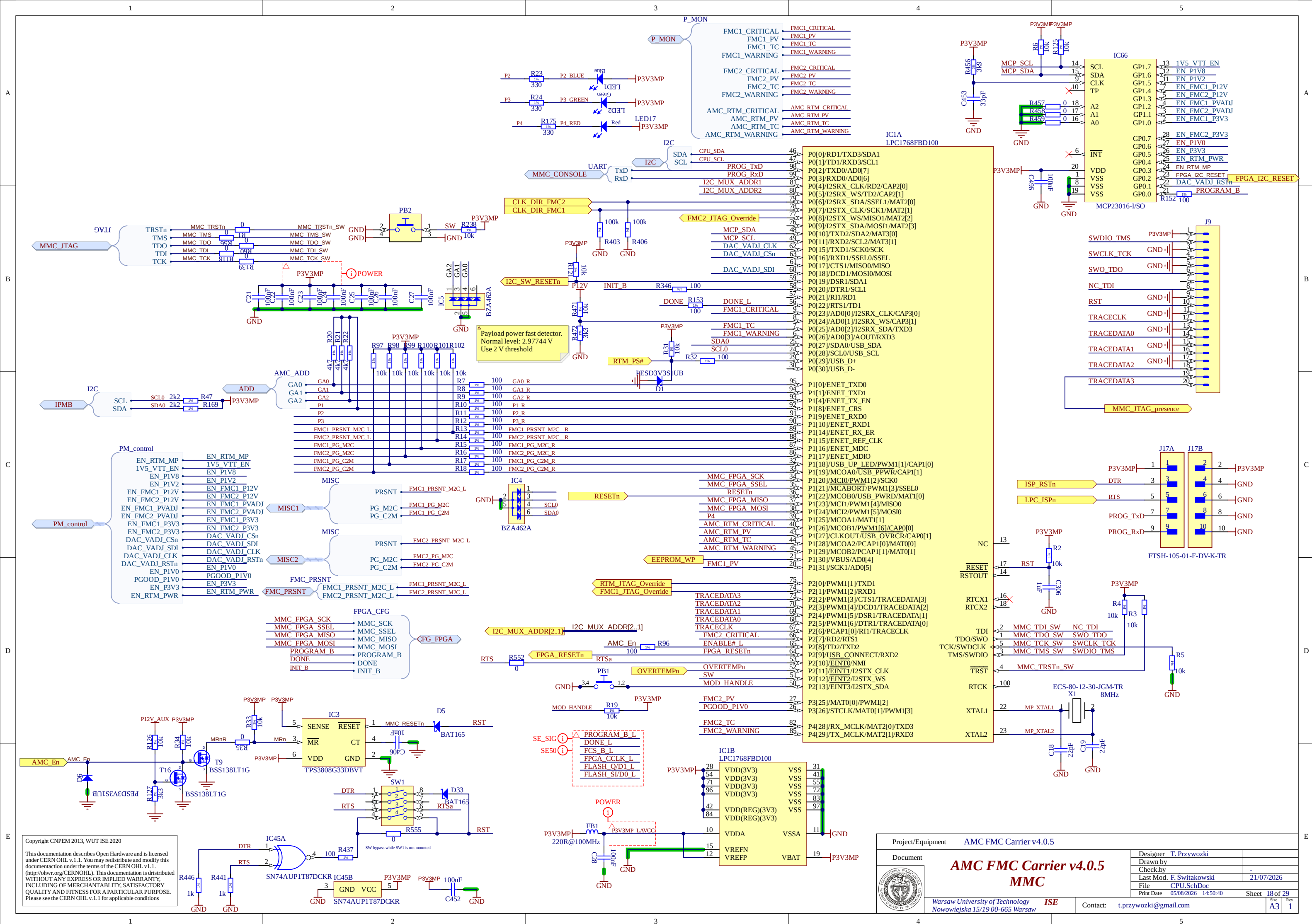
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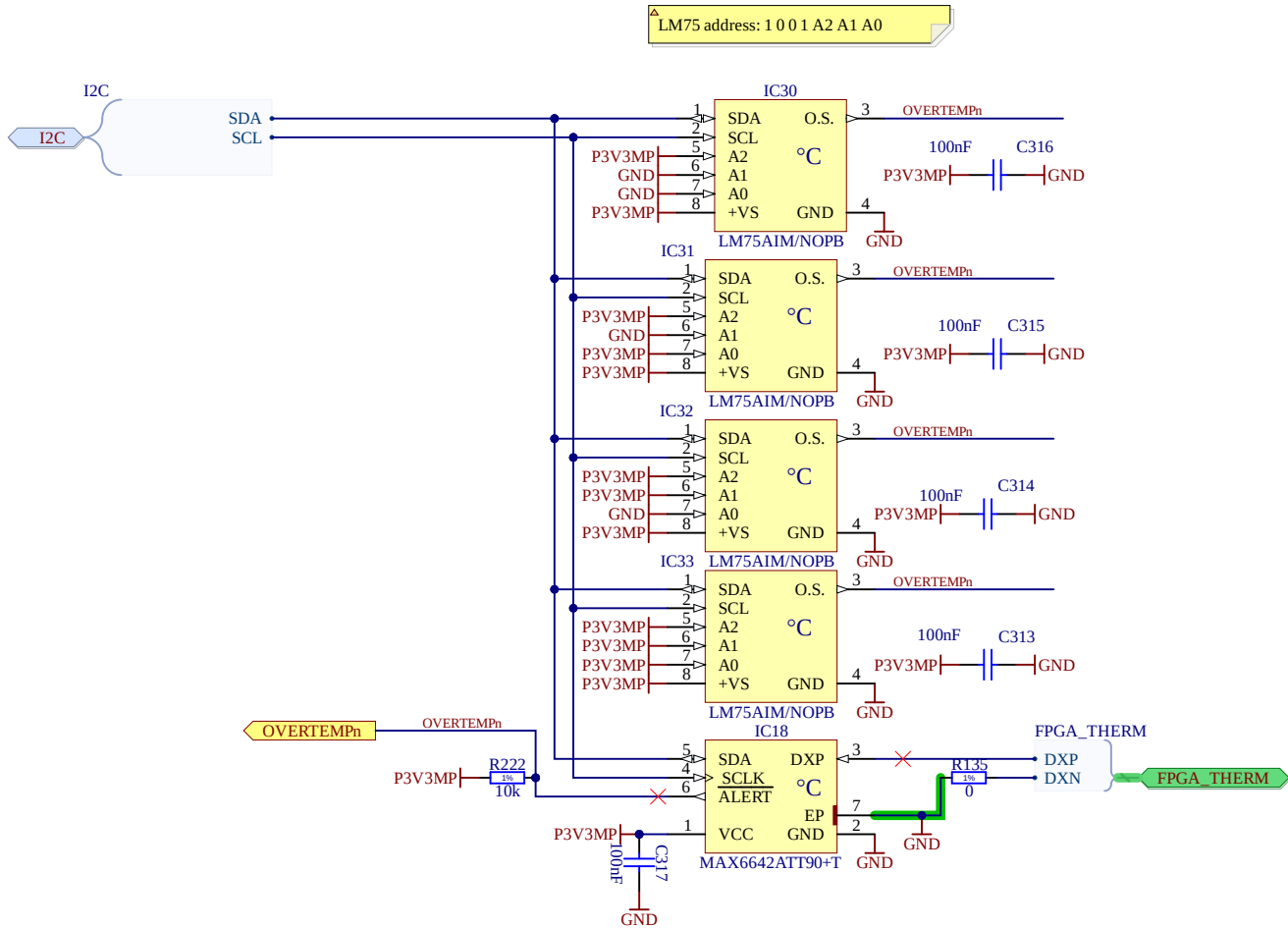


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Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		Contact:	t.przywozki@gmail.com

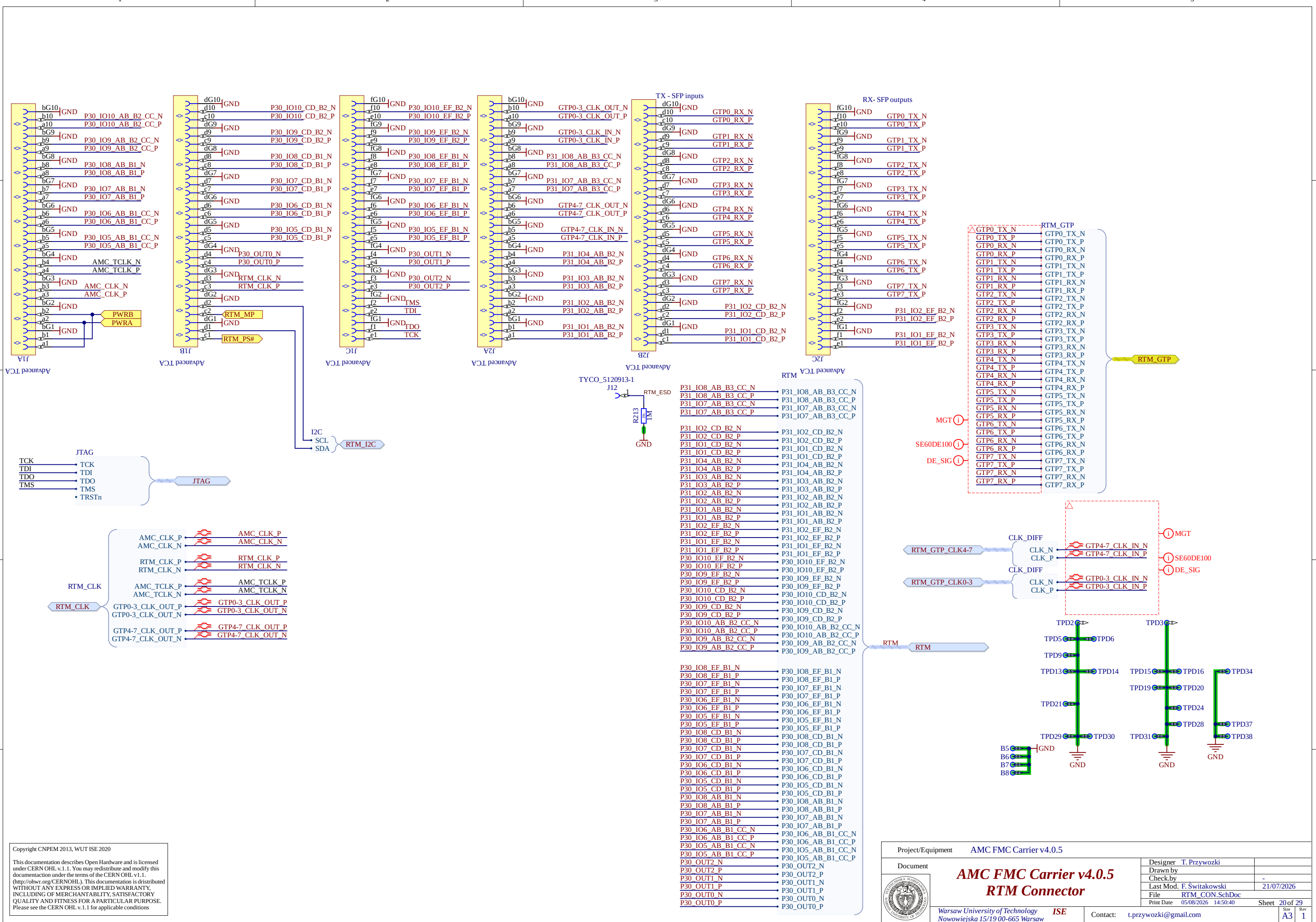




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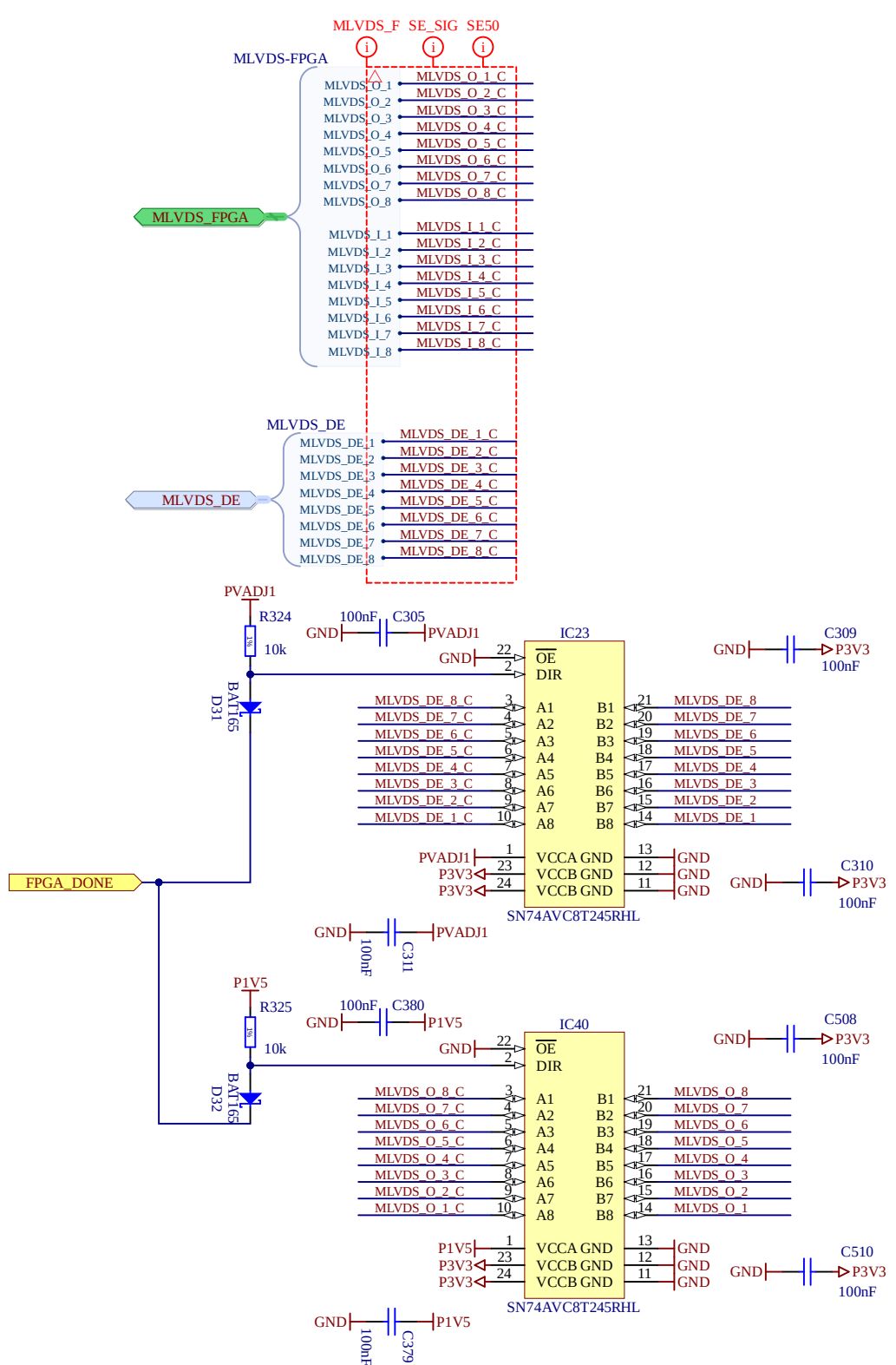


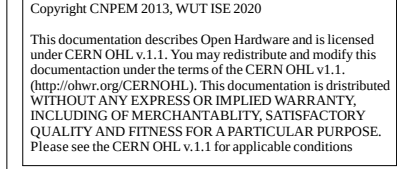
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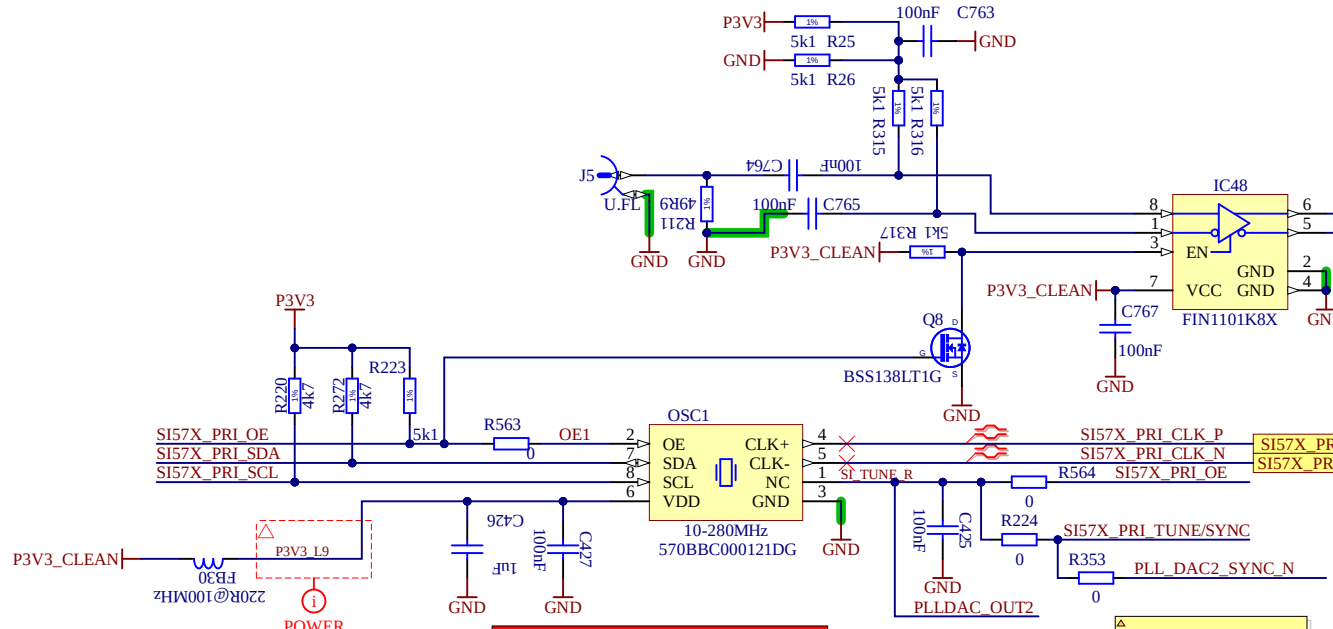
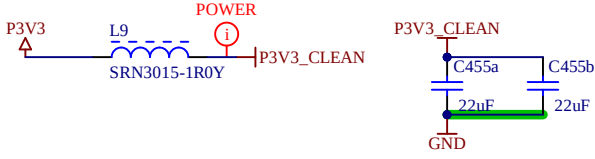
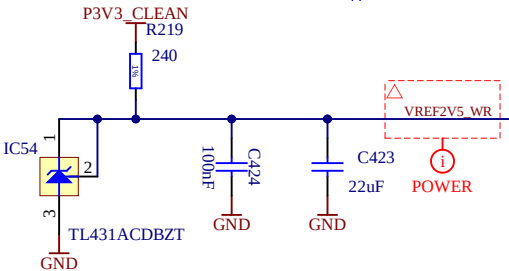
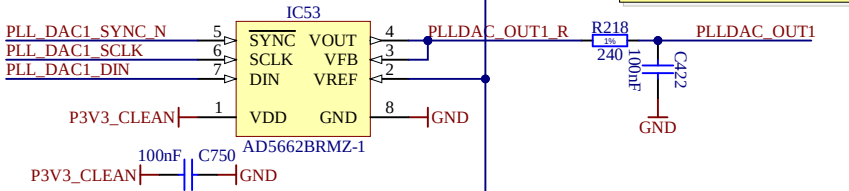
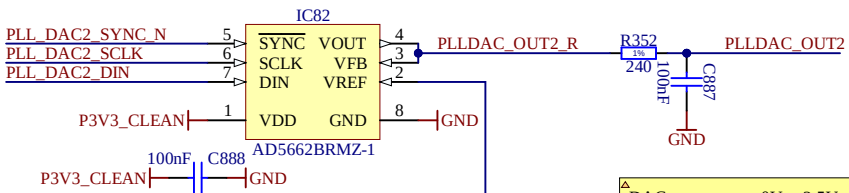
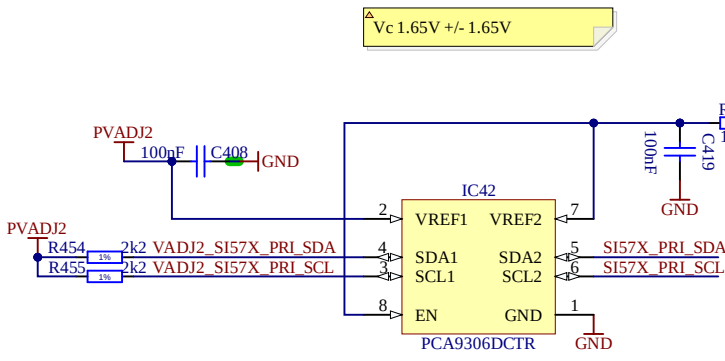
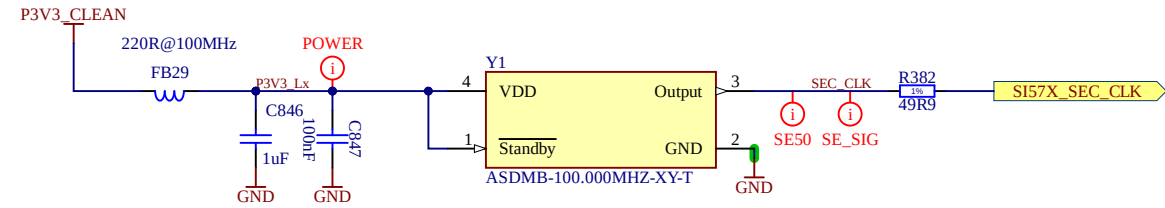
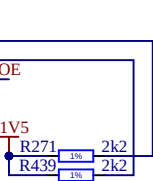
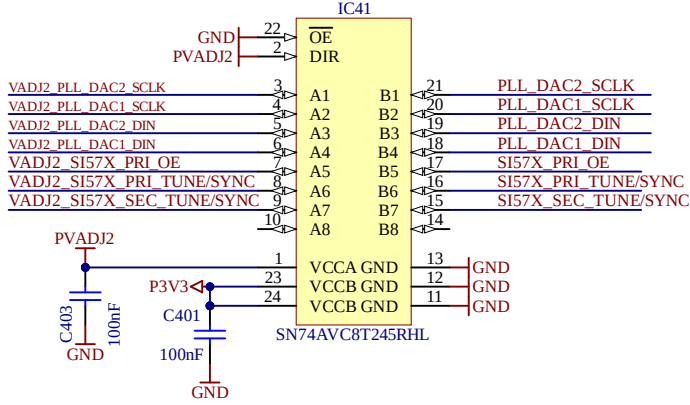
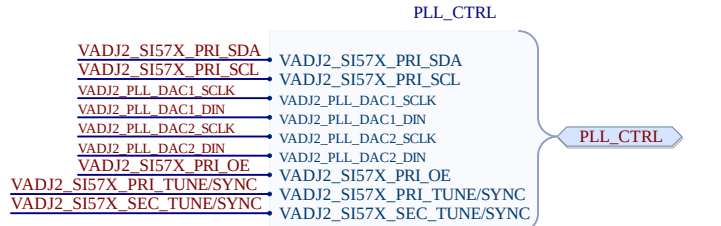
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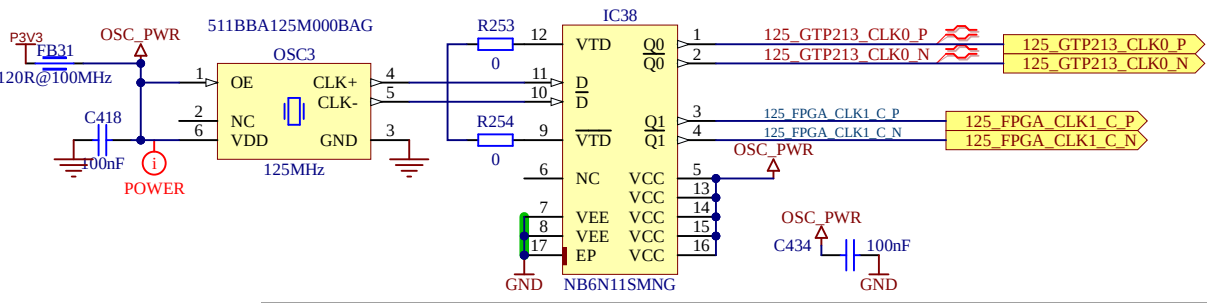
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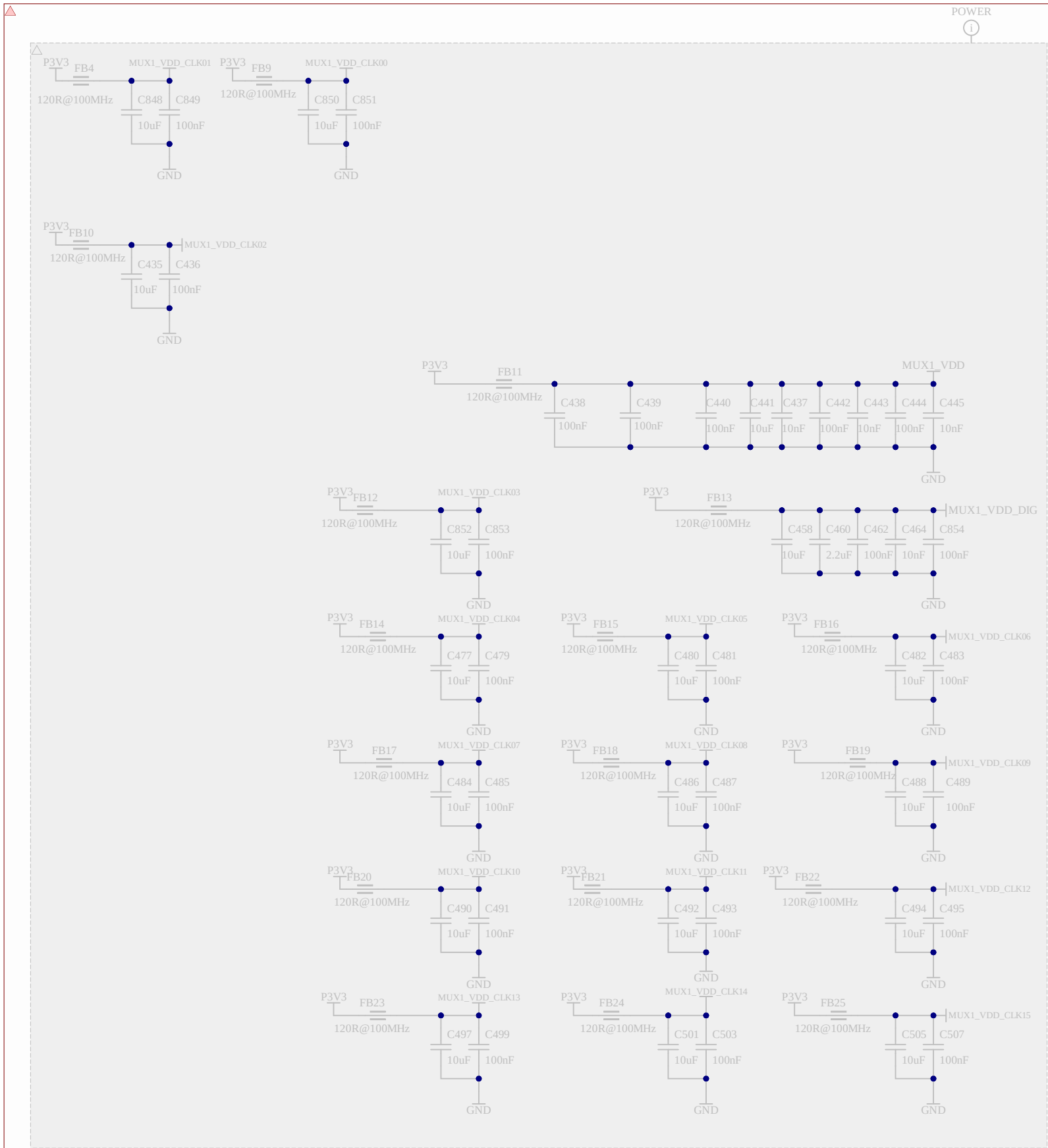
Check OSC Part Number in Variants!



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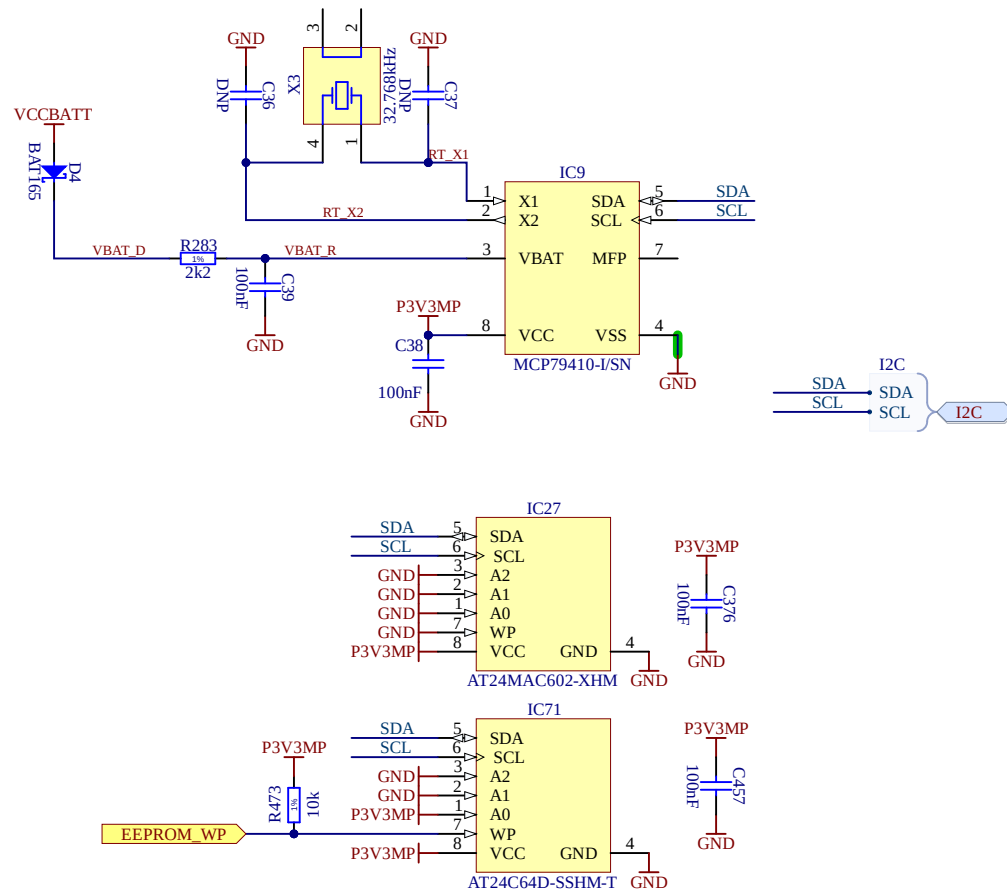
Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T. Przywozki	
		Drawn by	
		Check.by	
		Last Mod. F. Świątkowski	
		21/07/2026	
		File Oscillators.SchDoc	
		Print Date 05/08/2026 14:50:40	
		Sheet 23 of 29	
Warsaw University of Technology ISE		Contact: t.przywozki@gmail.com	Size A3 Rev 1



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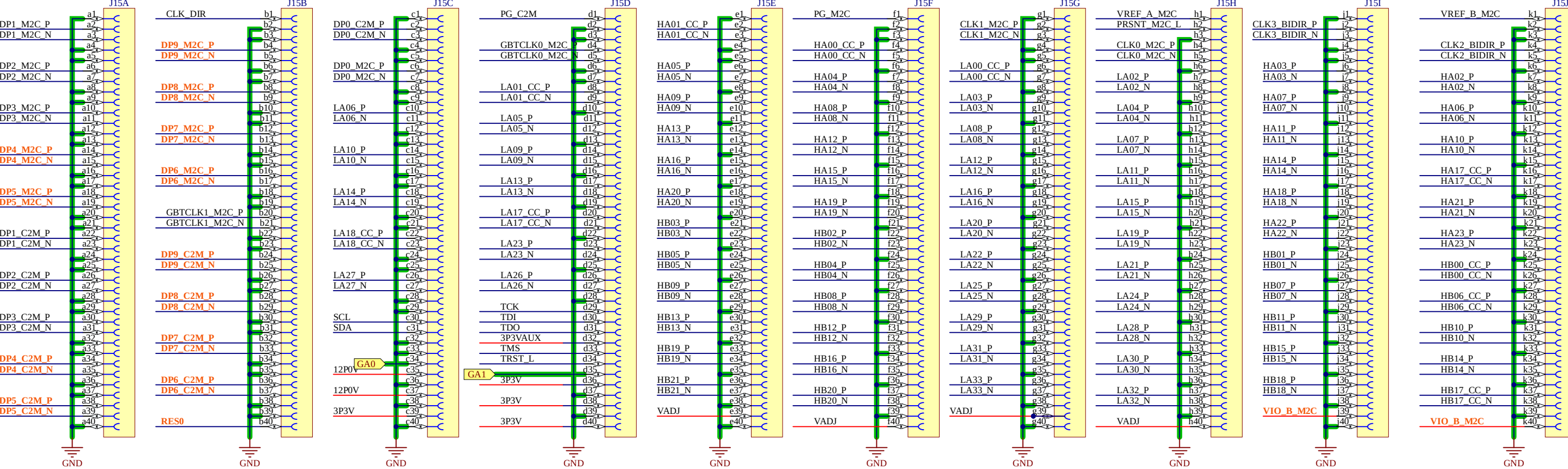
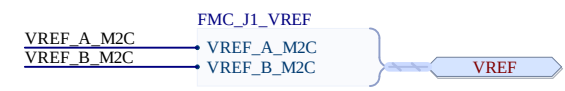
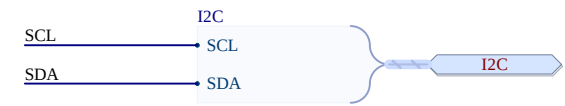
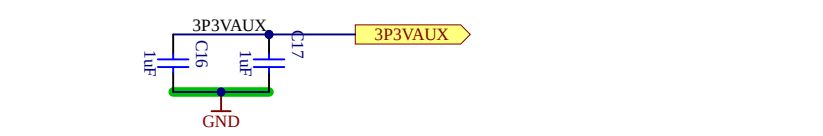
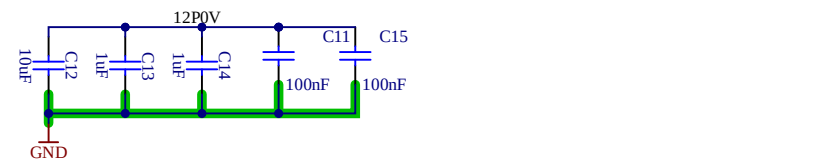
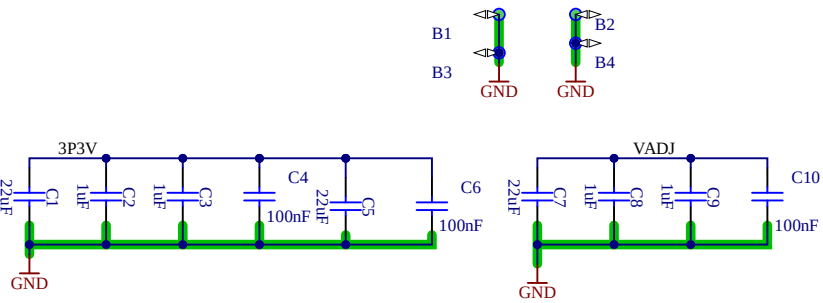
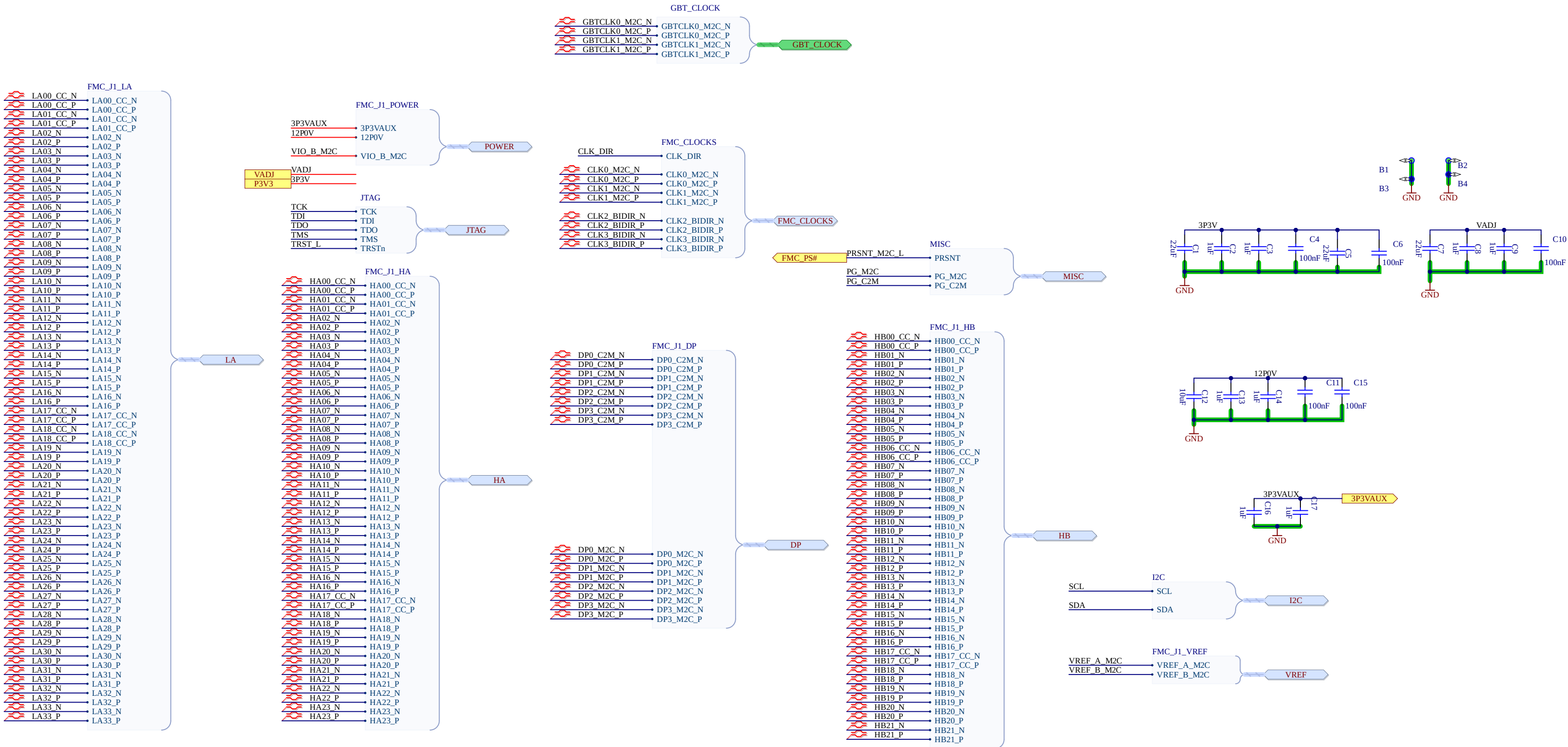
Project/Equipment		AMC FMC Carrier v4.0.5		
	Document	Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw ISE Contact: t.przywozki@gmail.com		
	AMC FMC Carrier v4.0.5 8V54816A Supply			
	Designer	T. Przywozki		
	Drawn by			
Check by	-			
Last Mod.	F. Świtakowski	21/07/2026		
File	8V54816A_SUPPLY.SchDoc			
Print Date	05/08/2026 14:50:40	Sheet 24 of 29		
		Size	Rev	
		A3	1	



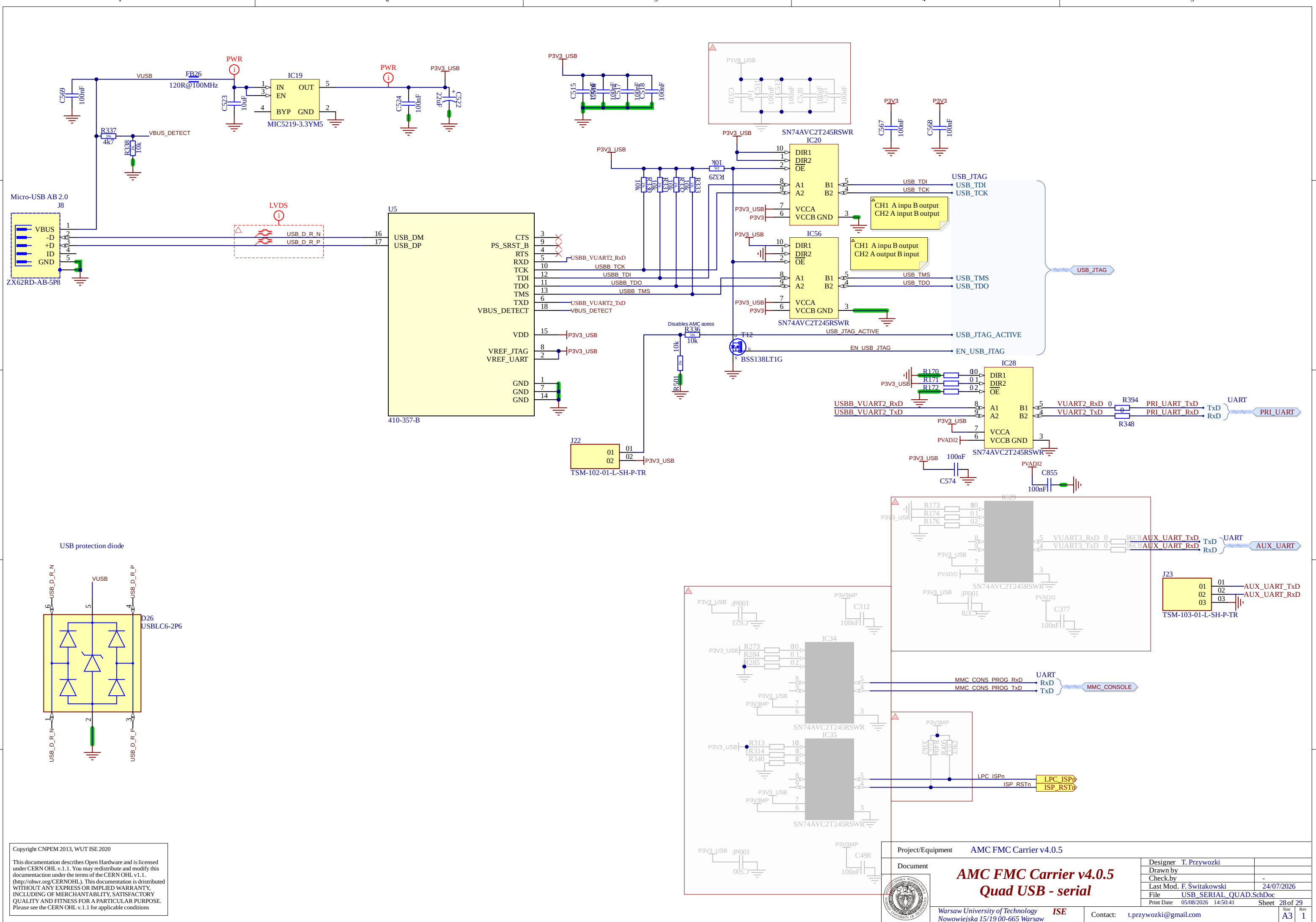
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	Document		Designer	T. Przywozki		
	AMC FMC Carrier v4.0.5 RTCE & EEPROM		Drawn by		XX/XX/XXXX	
			Check.by	-		
			Last Mod.	F. Świątkowski	21/07/2026	
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			Print Date	05/08/2026 14:50:40	Sheet	26 of 29
			Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		Contact:	t.przywozki@gmail.com
	ISE				Size A4	
				Rev 1		



VAdj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC



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A

B

C

D

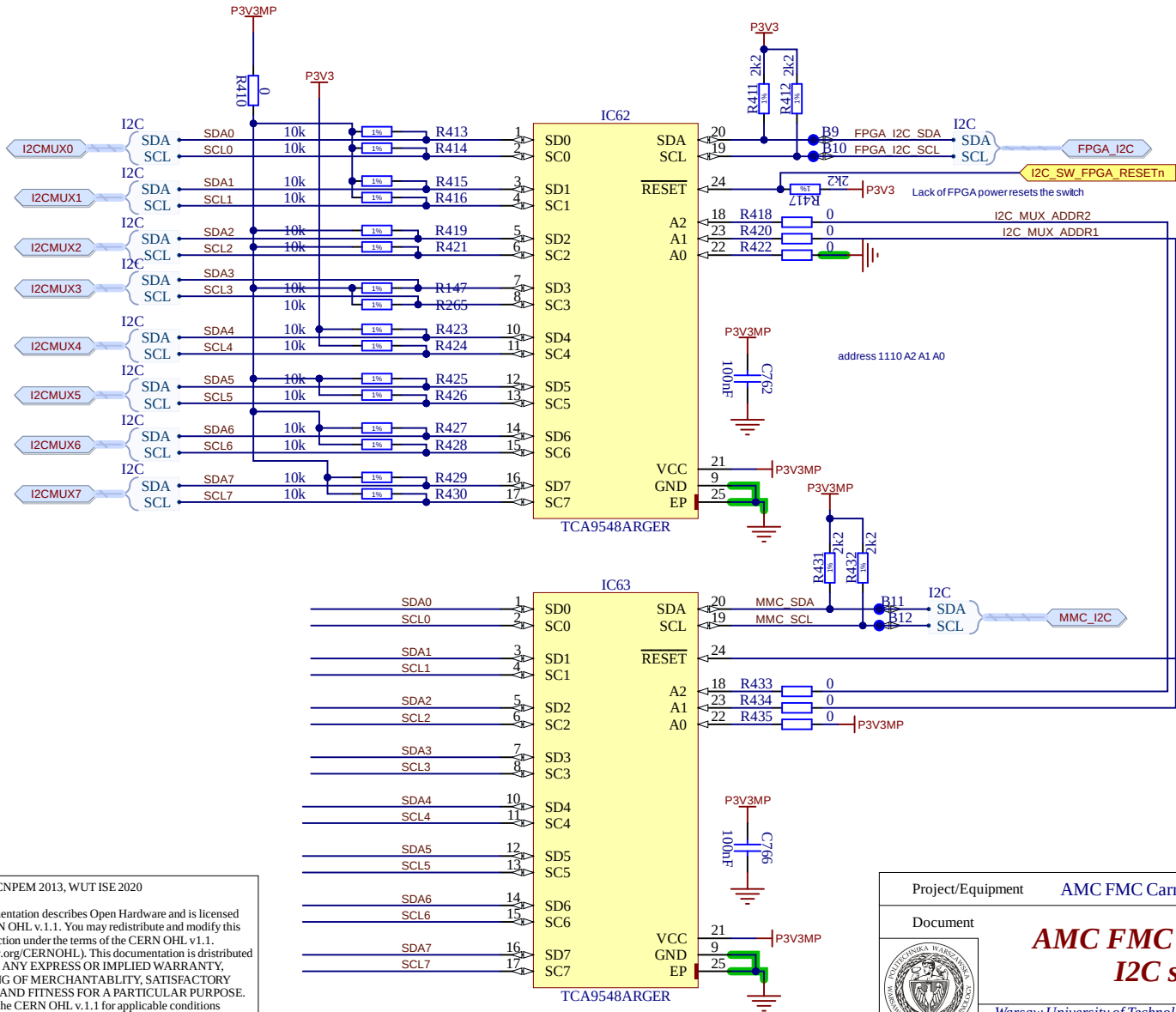
A

B

C

D

I2C switch footprint is compatible with MAX7358 which has interesting anti-lock capabilities



I2C device address map	MUX Port0	MAX6642 0x48
	Temp sensors	LM75: 0x4C LM75: 0x4D LM75: 0x4E LM75: 0x4F
	MUX Port1	RTC EEPROM: 0x57
	RTCE	SRAM/RTCC reg: 0x6F EEPROM 2k: 0x50 EUID: 0x59 EEPROM 64k: 0x51
	MUX Port2	
	MUX Port3	INA3221AIRGV: 0x40 Power
	MUX Port4	8V54816ANLG: 0x5B
	MUX Port5	RTM devices
	MUX Port6	EEPROM 0x52
	MUX Port7	EEPROM 0x50
	FMC1	

mux address 1110 A2 A1 0 from FPGA side 0x7x
mux address 1110 A2 A1 1 from CPU side 0x7x

I2C_MUX_ADDR[2..1] I2C_MUX_ADDR[2..1]

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Project/Equipment

AMC FMC Carrier v4.0.5

Document



AMC FMC Carrier v4.0.5
I2C switches

Designer

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